

CATALYSIS

A hydrophobic FeMn@Si catalyst increases olefins from syngas by suppressing C1 by-products

Yanfei Xu¹, Xiangyang Li², Junhu Gao², Jie Wang¹, Guangyuan Ma¹, Xiaodong Wen^{2,3}, Yong Yang^{2,3}, Yongwang Li^{2,3}, Mingyue Ding^{1*}

Although considerable efforts have been made in the selective conversion of syngas [carbon monoxide (CO) and hydrogen] to olefins through Fischer-Tropsch synthesis (FTS), ~50% of the converted CO is transformed into the undesired one-carbon molecule (C1) by-products [carbon dioxide (CO₂) and methane (CH₄)]. In this study, a core-shell FeMn@Si catalyst with excellent hydrophobicity was designed to hinder the formation of CO₂ and CH₄. The hydrophobic shell protected the iron carbide core from oxidation by water generated during FTS and shortened the retention of water on the catalyst surface, restraining the side reactions related to water. Furthermore, the electron transfer from manganese to iron atoms boosted olefin production and inhibited CH₄ formation. The multifunctional catalyst could suppress the total selectivity of CO₂ and CH₄ to less than 22.5% with an olefin yield of up to 36.6% at a CO conversion of 56.1%.

Olefins are feedstocks for many high-value-added chemical products and are usually produced from petroleum (1), so there is increasing interest in alternative production routes from nonpetroleum carbon resources (2), including syngas (a mixture of CO and H₂) that can be derived from biomass, coal, and natural gas (3, 4). Recently, many breakthroughs have been made in the conversion of syngas to olefins. Bao and colleagues (5, 6) reported the concept of oxide-zeolite (OX-ZEO), separating CO activation and C–C coupling on different active sites. At 400°C, ZnCrO_x/SAPO and MnO_x/SAPO exhibited a high selectivity of 80% for light olefins (C₂ to C₄) with CO conversion of ~17 and ~8%, respectively. Later, Wang and colleagues (7) prepared a ZnZrO_x/SAPO catalyst that had 70% selectivity for light olefins with ~10% CO conversion at 400°C. Although greater olefin selectivity in hydrocarbons was obtained through the OX-ZEO pathway, a lower catalytic activity and olefin yield were exhibited even at a very high reaction temperature.

Alternatively, another route, called Fischer-Tropsch to olefins (FTO), was developed for selectively converting syngas to olefins with a relatively high activity. Zhong *et al.* (8) reported that 61% selectivity for light olefins could be obtained over CoMn oxides with ~32% CO conversion. After de Jong and colleagues (9) found excellent selectivity of light olefins over iron-based catalysts for Fischer-Tropsch synthesis (FTS), research on Fe active metal ramped up (10–12). Liu *et al.* (13) prepared

Mn-modified Fe₃O₄ nanoparticles, which exhibited 60% selectivity for light olefins with ~42% CO conversion. Our previous work (14) indicated that a core-shell Fe₃O₄@MnO₂ catalyst presented a greater olefin selectivity of 79% with ~68% CO conversion.

The olefin selectivity reported above refers to the hydrocarbon products and does not include the CO₂ fraction. However, during the conversion process of syngas, the water-gas shift (WGS) reaction is inevitable given the coexistence of CO and H₂O under high reaction temperatures, which results in a high CO₂ selectivity (between 40 and 50%) when converting CO (5–9, 14, 15). Moreover, ~10 to 20% selectivity of by-product CH₄ is generally presented in the FTS reaction, especially for the iron-based catalysts (9, 10, 13, 16). Thus, during the syngas-to-olefin reaction, >45% of the converted CO is transformed into the undesired one-carbon molecule (C1) by-products (CO₂ and CH₄) either through the OX-ZEO process (5) or the FTO path (8, 9), as depicted in Fig. 1A. Recently, some work has been done to try to address this issue (17, 18), but the total selectivity of C1 products was still very high and the olefin yield was very low. Notably, Wang *et al.* (19) and Xie *et al.* (20) reported the suppression of C1 products by a specific iron carbide phase and by cobalt promoted with Mn/Na/S, respectively. However, these catalysts could only run at relatively low CO conversion levels (<20%), implying poor olefin yield.

We present a core-shell FeMn@Si catalyst with excellent hydrophobicity that hindered the WGS reaction, which enhanced the C–C coupling reaction and inhibited the formation of CO₂ and CH₄. The chain growth probability was increased to 0.73 (fig. S1), and the selectivity for undesired C1 by-products was suppressed to 22.1% (Fig. 1A). Moreover, the multifunctional catalyst exhibited an olefin selectivity of 83.3% (including 68.2%

of α -olefins) and an olefins/paraffins molar ratio of 5.0 in the C₂₊ hydrocarbons (Fig. 1B). Reaction conditions, such as reaction temperature, space velocity, and pressure, were tuned to modify the catalytic performance (figs. S2 to S4). Under the optimum reaction conditions of 320°C, 3.0 MPa, 4000 ml·h^{−1}·g^{−1}, H₂/CO = 2, the yield of olefins reached as high as 36.6% at a CO conversion of 56.1% in a single run (table S1).

The stability of catalysts is another crucial index to advance industrial applications. In this study, the CO conversion and product distribution remained stable with time on stream (Fig. 1C). The selectivity of undesired C1 by-products was suppressed to ~23% and the olefin selectivity in total products remained at ~65% during the 100-hour stability test. All of the above catalytic tests suggested that our catalyst worked well for efficiently converting syngas to olefins while successfully suppressing the formation of undesired CO₂ and CH₄.

We used several characterization methods to study the origin of the low C1 selectivity and high olefin yield. An unpromoted Fe₂O₃@SiO₂ catalyst was used to investigate the formation path and inhibition mechanism of CO₂ during iron-based FTS reaction. An appropriate amount of Mn promoter was then added to enhance the C–C coupling reaction, which suppressed the CH₄ selectivity and boosted olefin production while maintaining low CO₂ selectivity. The core-shell Fe₂O₃@SiO₂ (Fe@Si) was prepared using the modified Stöber method with nanorod-shaped Fe₂O₃ (figs. S5 and S6). Transmission electron microscopy (TEM) images directly confirmed that the nanorod-shaped Fe₂O₃ was encapsulated by a homogeneous SiO₂ shell (Fig. 2A and fig. S7). As shown in the Fourier transform infrared (FTIR) spectra (Fig. 2C), the absorption bands at 465 and 542 cm^{−1} were assigned to the vibration of Fe–O bonds in the Fe₂O₃ core (21), and the bands at 471, 800, and 1095 cm^{−1} to the vibration of Si–O–Si bonds in the SiO₂ shell. Additionally, an absorption band appeared at 950 cm^{−1}, corresponding to the stretching vibration of Si–OH (22). The OH stretch of isolated silanol groups at 3740 cm^{−1} on Fe@Si was detected in the diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) (fig. S8), providing the possibility to introduce hydrophobic –CH₃ groups through the silylation reaction (fig. S9).

The obtained Fe@Si was rendered hydrophobic (Fe@Si-c) by further organic modification with chlorotrimethylsilane (TMCS). This modification had no obvious influence on the morphology, phase composition, or pore structure of Fe@Si (figs. S10 to S12). TEM images of Fe@Si-c showed that the nanorod-shaped Fe₂O₃ was also well encapsulated by the amorphous SiO₂ shell (Fig. 2B). FTIR and DRIFTS revealed new absorption bands at ~2925 cm^{−1} for Fe@Si-c compared with Fe@Si (Fig. 2C and fig. S8), which were attributed

¹School of Power and Mechanical Engineering, Wuhan University, Wuhan 430072, China. ²Synfuels China Co., Ltd., Beijing 101407, China. ³State Key Laboratory of Coal Conversion, Institute of Coal Chemistry, Chinese Academy of Sciences, Taiyuan 030001, China.

*Corresponding author. Email: dingmy@whu.edu.cn (M.D.)

to the vibration of introduced $-\text{CH}_3$ groups (22). The water-droplet contact angle tests were also conducted to characterize the hydrophobicity of Fe@Si and Fe@Si-c. Fe@Si exhibited a contact angle of 40.2° , implying a hydrophilic surface (Fig. 2E). After the hydrophobic modification, the contact angle increased to 137.6° over the Fe@Si-c surface, indicating that hydrophilic Fe@Si had been completely converted to the hydrophobic surface of Fe@Si-c through TMCS modification. Although Fe@Si was well dispersed in water, it was difficult to infiltrate the hydrophobic surface of Fe@Si-c with water, and it formed a long-lasting separate phase (Fig. 2F). The water vapor adsorption and desorption tests also confirmed that the hydrophobic modification hindered the adsorption of water molecules (fig. S13).

The hydrophobic shell coated on iron species played a critical role in tuning the active iron phases during FTS. Compared with fresh Fe@Si, new absorption bands at 578 and 700 cm^{-1} were detected in the FTIR spectra of spent Fe@Si (Fig. 2D) and were related to the vibration of the Fe-O bond in the Fe_3O_4 phase (23); bands attributed to the vibration of the Fe-O bond in the Fe_2O_3 phase disappeared. Furthermore, both Fe_3O_4 and iron carbides were detected in the x-ray diffraction (XRD) pattern of spent Fe@Si (Fig. 3A), which was consistent with the Fe_2O_3 phase in traditional iron-based catalysts being converted into a mixture of iron oxides and carbides with reaction time on stream (4, 24, 25).

For Fe@Si-c, no absorption bands related to the vibration of Fe-O bonds in Fe_2O_3 and Fe_3O_4 were seen in the FTIR spectra after reaction (Fig. 2D), implying that iron oxides disappeared during FTS. Only iron carbides were detected in the XRD pattern of spent Fe@Si-c (Fig. 3A), demonstrating that iron oxides were completely converted to carbides during FTS. We also used Mössbauer spectra to determine the content of the iron phases. The spent Fe@Si consisted of 60.7% Fe_3O_4 and 39.3% iron carbides ($\chi\text{-Fe}_5\text{C}_2$ and $\theta\text{-Fe}_3\text{C}$), whereas the spent Fe@Si-c consisted only of iron carbides (Fig. 3, B and C, and table S2), directly confirming that the hydrophobic shell could effectively tune iron phases during reaction.

To study the phase transformation in more detail, we synthesized iron carbides by introducing a carburization procedure (320°C , 0.1 MPa, $4000\text{ ml}\cdot\text{h}^{-1}\cdot\text{g}^{-1}$, $\text{H}_2/\text{CO} = 1$, 5 hours) and compared phase composition of catalysts during FTS. After carburization, the Fe_2O_3 phase in both Fe@Si and Fe@Si-c was completely transformed into iron carbides (Fig. 3D, fig. S14, and table S3). During the subsequent FTS reaction, iron carbides inside Fe@Si were converted into a mixture of iron carbides and Fe_3O_4 phases, whereas Fe@Si-c was only com-

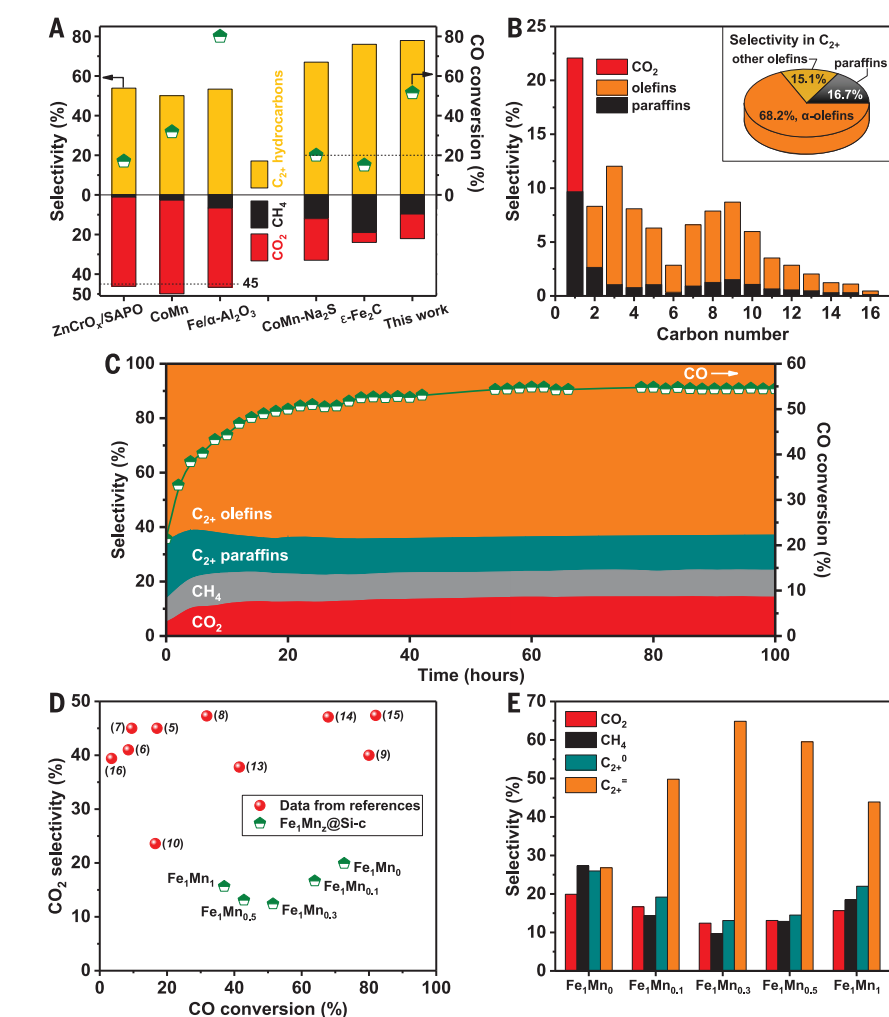


Fig. 1. Catalytic performance for syngas to olefins. (A) Product distribution (including CO₂) of FeMn@Si-c in comparison with previous work (5, 8, 9, 19, 20). **(B)** Detailed product distribution (including CO₂) and the selectivity of olefins in C₂₊ hydrocarbons (inset) over FeMn@Si-c. **(C)** A stability test of FeMn@Si-c. **(D)** CO conversion and CO₂ selectivity and **(E)** product distribution (including CO₂) over Fe₁Mn₂@Si-c with different Mn/Fe ratios. Reaction conditions: 320°C , 2.0 MPa, $4000\text{ ml}\cdot\text{h}^{-1}\cdot\text{g}^{-1}$, $\text{H}_2/\text{CO} = 2$.

posed of iron carbides (Fig. 3D, fig. S15, and table S4). Given that the only difference between Fe@Si and Fe@Si-c was their water-droplet contact angle, the different phase evolution processes on the two catalysts could be attributed to the hydrophobicity of the catalyst surface. As the FTS reaction proceeded, water was produced and the surrounding ambient became more oxidizing. As a result, iron carbides were partially converted into Fe_3O_4 by water oxidation (26–28), leading to the coexistence of iron oxides and carbides in Fe@Si. The hydrophobic surface on Fe@Si-c would facilitate rapid water desorption and inhibit its entry into the catalyst. Thus, although the FTS reaction produced water, its restricted entry inhibited the conversion of iron carbides back to oxides.

It is generally accepted that CO₂ formation on iron-based FTS catalysts consists of pri-

mary CO₂ and secondary CO₂ (19, 29–32). In the CO₂ selectivity–CO conversion curve, the extrapolated CO₂ selectivity toward the axis intercept was ascribed to the primary CO₂ ($\text{CO}^* + \text{O}^* \rightarrow 1\text{ CO}_2$, where the asterisk means chemisorbed). The increasing CO₂ selectivity above the axis intercept was assigned to the secondary CO₂, which was related to the re-adsorption of water and its subsequent WGS reaction with CO. The slope of the curve reflected the rate of secondary CO₂ formation stemming from the WGS reaction. The CO₂ selectivity–CO conversion curve (Fig. 4A) was obtained by adjusting the flow rate at a constant H_2/CO ratio. Fe@Si exhibited a primary CO₂ selectivity of ~10%, whereas Fe@Si-c was free of primary CO₂. Because Fe@Si was composed of a mixture of Fe_3O_4 and iron carbides, whereas Fe@Si-c only consisted of iron carbides during reaction (Fig. 3 and fig. S16), we

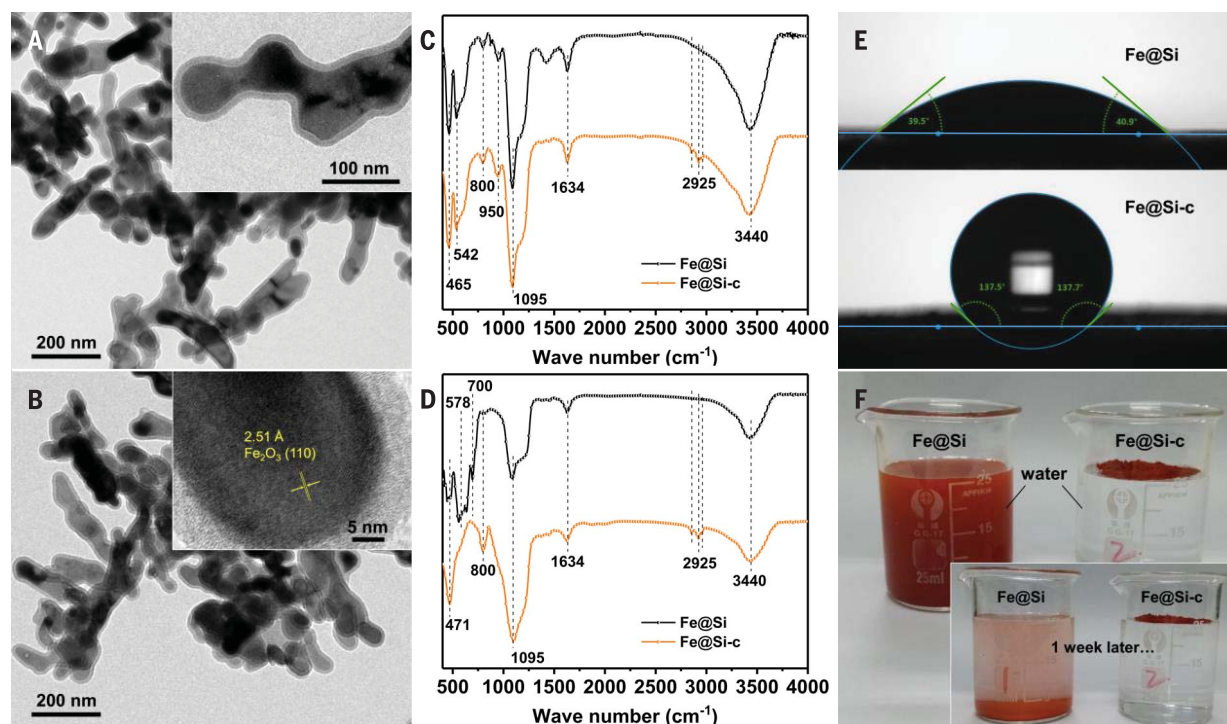


Fig. 2. Structural characterization of the Fe@Si and Fe@Si-c catalysts. (A and B) TEM images of Fe@Si (A) and Fe@Si-c (B). (C and D) FTIR spectra of the fresh (C) and spent (D) catalysts. (E) Water-droplet contact angle tests. (F) Photographs of catalysts dispersed in water, initially and 1 week later.

attributed primary CO₂ formation to the Fe₃O₄ presented in the Fe@Si catalyst, consistent with the previous report (19).

The slope of the curve obtained over Fe@Si was 1.81 times that obtained over Fe@Si-c (Fig. 4A), suggesting that Fe@Si favored the formation of secondary CO₂. However, the hydrophobic shell coated on Fe@Si-c facilitated the timely desorption of water and prevented the readsorption of water, reducing the local water partial pressure in the vicinity of iron species and inhibiting the WGS reaction. Also, the Fe₃O₄ phase is the most active phase for the WGS reaction in the iron-based FTS catalysts (19, 29, 33, 34) and would form more by-product CO₂ from CO (figs. S17 and S18 and table S5). The hydrophobic shell coated on Fe@Si-c inhibited the reoxygenation of iron carbides toward Fe₃O₄, further inhibiting the WGS reaction. A >40% reduction in CO₂ emission was achieved at a medium CO conversion level (~41%) (Fig. 4B).

To certify the above conclusions, the Fe@Si-c catalysts with different hydrophobic abilities were prepared by changing the TMCS coverage. As shown in fig. S19, the increase in the TMCS exposure increased the water contact angle from 40.2° for Fe@Si to 125.9° for Fe@Si-c_{0.5} (0.5 ml per gram of catalyst) and to 137.6° for Fe@Si-c_{2.0}, implying gradually enhanced hydrophobicity. Moreover, the slope of curve obtained over Fe@Si-c_{0.5} was 1.26 times that obtained over Fe@Si-c_{2.0} (fig. S20), further confirming that the enhancement of

catalyst hydrophobicity could suppress the WGS reaction by inhibiting the readsorption of water. For TMCS exposures >2.0 ml/g, the water contact angle and the slope of the curve tended to remain stable (figs. S19 and S20), suggesting that excessive addition of TMCS had no obvious effect on the catalyst hydrophobicity and the suppression of the WGS reaction. Fe@Si-c_{2.0} was chosen as the appropriate catalyst for the work that follows unless otherwise noted.

We tuned the H₂/CO ratio in syngas to modify the CO₂ selectivity of Fe@Si-c. As shown in fig. S21A, upon increasing the H₂/CO ratio in syngas from 1 to 4, CO conversion gradually increased from 55.5 to 80.2%, accompanied by a decrease in CO₂ selectivity from 35.0 to 15.0%, demonstrating that a higher H₂/CO ratio increased the amount of H₂ in the system and suppressed the WGS reaction. However, the selectivity of CH₄ (the other undesired C1 by-product) in hydrocarbons increased from 14.1 to 47.6% because of enhanced hydrogenation (fig. S21B).

The formation path and inhibition mechanism of CO₂, on the basis of the above results, were illustrated in fig. S22. During the FTS reaction, the adsorbed H₂* and CO* on the surface of iron species reacted to form surface CH_x species, and further C-C coupling, dehydrogenation, and hydrogenation reactions would produce various olefins and paraffins. The chemisorbed O* formed by CO dissociation could be removed by reacting with co-

adsorbed H* and CO* to form H₂O and CO₂, respectively, as oxygen carriers through primary FTS steps. Readsorption of H₂O on the surface of iron species and reaction with CO* would form secondary CO₂ through the WGS reaction. Phase transformation of iron oxides to iron carbides would remove O* from the iron oxides and form primary CO₂. During reaction, Fe@Si-c consisted only of iron carbides, which were free of primary CO₂. The inhibited adsorption of H₂O and the absence of Fe₃O₄ further suppressed the WGS reaction, so the CO₂ selectivity over the Fe@Si-c catalyst during FTS was obviously reduced. Inhibition of the WGS reaction on Fe@Si-c would decrease the amount of H₂ produced and suppress over-hydrogenation reaction, promoting olefins over CH₄ (fig. S23). The olefin selectivity in hydrocarbons increased from 11.1% of Fe@Si to 65.3% of Fe@Si-c.

Mn as a promoter is widely used to boost olefin formation in FTS. We prepared and characterized the Mn-modified catalysts (figs. S24 to S26). Fe@Si-c with moderate addition of Mn (Mn/Fe ratio ≤ 0.3) still suppressed water adsorption and inhibited the oxidation of iron carbides during reaction. However, as the Mn/Fe ratio exceeded 0.3, both Fe₃O₄ and iron carbide phases were detected in the spent Fe₁Mn_x@Si-c catalyst (figs. S27 and S28 and table S6), because excessive Mn hindered the effective contact between the hydrophobic shell and iron carbide core (fig. S29) and led to partial oxidation.

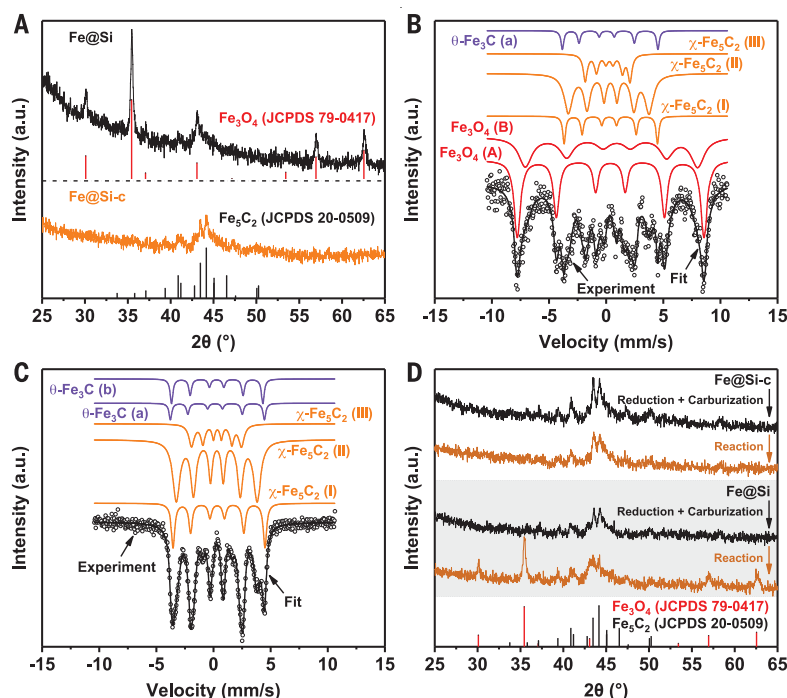


Fig. 3. Comparison of phase composition. (A) XRD patterns of the spent catalysts. (B and C) Mössbauer spectra of the spent Fe@Si (B) and Fe@Si-c (C). (D) XRD patterns of the catalysts after reduction and carburization and subsequent reaction. Reaction conditions: 320°C, 2.0 MPa, H₂/CO = 1. a.u., arbitrary units.

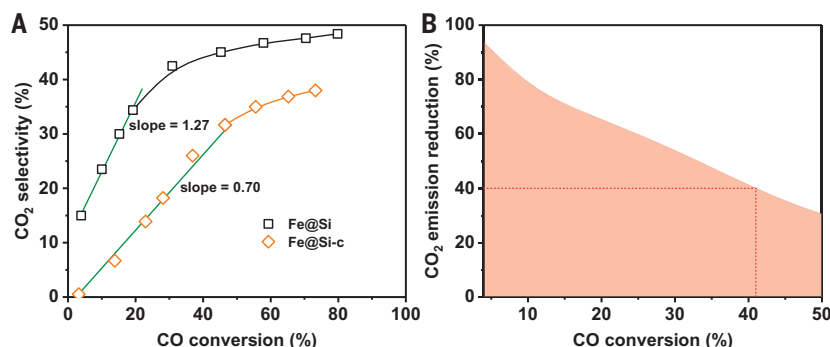


Fig. 4. Effect of hydrophobic modification on the performance of catalysts. (A) CO₂ selectivity–CO conversion curve. Reaction conditions: 320°C, 2.0 MPa, H₂/CO = 1. (B) CO₂ emission reduction of Fe@Si-c in comparison with Fe@Si. A >40% reduction in CO₂ emission was achieved at a medium CO conversion level (~41%).

The catalytic performance of Fe₁Mn_{0.3}@Si-c catalysts was shown in Fig. 1, D and E. The CO₂ selectivity obtained over all the Fe₁Mn_{0.3}@Si-c catalysts was suppressed to a low level (<20%) (Fig. 1D and fig. S30), much lower than the reported data under all the CO conversion levels in the previous references (5–10, 13–16), suggesting that the WGS reaction was still inhibited after the addition of Mn. Notably, an inverse volcano curve between CO₂ selectivity and CO conversion was observed over the Fe₁Mn_{0.3}@Si-c catalysts. CO₂ selectivity gradually decreased with the increase of Mn content and reached the minimum value of 12.4% with 51.5% CO conversion over Fe₁Mn_{0.3}@Si-c.

More importantly, moderate addition of Mn promoted the formation of olefins and suppressed that of CH₄. The olefin selectivity in total products increased with the increasing Mn content and reached the maximum value of 64.9% over Fe₁Mn_{0.3}@Si-c (Fig. 1E), which we attributed to the enhanced electron transfer from Mn to Fe atoms (fig. S31). This process facilitated the dissociated adsorption of CO and weakened hydrogenation in favor of the C–C coupling reaction and olefin production (14). Further increase of Mn content and formation of Fe₃O₄ during FTS hindered the effective contact between the hydrophobic shell and the iron species core (fig. S29), which

unproductively enhanced the WGS reaction and promoted the formation of C1 by-products (Fig. 1E and fig. S32).

REFERENCES AND NOTES

- H. M. Torres Galvis, K. P. de Jong, *ACS Catal.* **3**, 2130–2149 (2013).
- W. Zhou et al., *Chem. Soc. Rev.* **48**, 3193–3228 (2019).
- Q. H. Zhang, J. C. Kang, Y. Wang, *ChemCatChem* **2**, 1030–1058 (2010).
- J. Bao, G. H. Yang, Y. Yoneyama, N. Tsubaki, *ACS Catal.* **9**, 3026–3053 (2019).
- F. Jiao et al., *Science* **351**, 1065–1068 (2016).
- Y. F. Zhu et al., *ACS Catal.* **7**, 2800–2804 (2017).
- K. Cheng et al., *Angew. Chem. Int. Ed.* **55**, 4725–4728 (2016).
- L. Zhong et al., *Nature* **538**, 84–87 (2016).
- H. M. Torres Galvis et al., *Science* **335**, 835–838 (2012).
- J. Z. Lu et al., *ACS Catal.* **4**, 613–621 (2014).
- P. Zhai et al., *Angew. Chem. Int. Ed.* **55**, 9902–9907 (2016).
- X. Chen, D. Deng, X. Pan, Y. Hu, X. Bao, *Chem. Commun.* **51**, 217–220 (2015).
- Y. Liu, J. F. Chen, J. Bao, Y. Zhang, *ACS Catal.* **5**, 3905–3909 (2015).
- J. Wang et al., *ACS Appl. Mater. Interfaces* **10**, 43578–43587 (2018).
- V. P. Santos et al., *Nat. Commun.* **6**, 6451 (2015).
- O. Zhuo et al., *Chem. Sci.* **10**, 6083–6090 (2019).
- E. Rytter et al., *Catal. Today* **299**, 20–27 (2018).
- X. F. Yu et al., *Appl. Catal. B* **232**, 420–428 (2018).
- P. Wang et al., *Sci. Adv.* **4**, eaau2947 (2018).
- J. Xie et al., *Nat. Commun.* **10**, 167 (2019).
- S. M. Rodolfo-Baechler et al., *Mater. Lett.* **58**, 2447–2450 (2004).
- L. H. Shi, D. B. Li, B. Hou, Y. H. Sun, *Chin. J. Catal.* **28**, 999–1002 (2007).
- Y.-S. Li, J. S. Church, A. L. Woodhead, *J. Magn. Magn. Mater.* **324**, 1543–1550 (2012).
- E. de Smit, B. M. Weckhuysen, *Chem. Soc. Rev.* **37**, 2758–2781 (2008).
- M. Y. Ding et al., *Appl. Energy* **160**, 982–989 (2015).
- C. N. Satterfield, R. T. Hanlon, S. E. Tung, Z. M. Zou, G. C. Papaefthymiou, *Ind. Eng. Chem. Prod. Res. Dev.* **25**, 407–414 (1986).
- D. B. Bukur, B. Todir, N. Elbasher, *Catal. Today* **275**, 66–75 (2016).
- J. H. Liu et al., *Ind. Eng. Chem. Res.* **57**, 9120–9126 (2018).
- Y. Cheng et al., *Appl. Catal. B* **204**, 475–485 (2017).
- S. Krishnamoorthy, A. Li, E. Iglesia, *Catal. Lett.* **80**, 77–86 (2002).
- S. Li, S. Krishnamoorthy, A. Li, G. D. Meitzner, E. Iglesia, *J. Catal.* **206**, 202–217 (2002).
- M. Ojeda et al., *J. Catal.* **272**, 287–297 (2010).
- C. Pirola, C. L. Bianchi, A. Di Michele, S. Vitali, V. Ragaini, *Catal. Commun.* **10**, 823–827 (2009).
- G. P. van der Laan, A. A. C. M. Beenackers, *Appl. Catal. A Gen.* **193**, 39–53 (2000).

ACKNOWLEDGMENTS

Funding: This work was supported by the National Natural Science Foundation of China (51861145102 and 21978225), National Key Research and Development Plan of China (2018YFE0125200), Guangdong Provincial Science and Technology Project (2019A050510031), Science and Technology program of Shenzhen (JCYJ20180302153928437), and Fundamental Research Fund for the Central Universities (2042019kf0221). **Author contributions:** Y.X. performed the catalyst preparation, characterization, and catalytic tests. X.L. and J.G. provided helpful characterization and analysis. J.W. and G.M. participated in the catalytic tests. X.W., Y.Y., and Y.L. provided helpful discussion. M.D. and Y.X. designed the study, analyzed the data, and wrote the paper. All authors approved the final version of the manuscript. **Competing interests:** A Chinese patent application held by Wuhan University that covers the preparation of catalysts is pending. **Data and materials availability:** All data are available in the manuscript or the supplementary materials.

SUPPLEMENTARY MATERIALS

science.sciencemag.org/content/371/6529/610/suppl/DC1
Materials and Methods
Figs. S1 to S32
Tables S1 to S6
References (35–39)

18 February 2020; accepted 18 December 2020
10.1126/science.abb3649



Supplementary Materials for
**A hydrophobic FeMn@Si catalyst increases olefins from syngas
by suppressing C1 by-products**

Yanfei Xu, Xiangyang Li, Junhu Gao, Jie Wang, Guangyuan Ma,
Xiaodong Wen, Yong Yang, Yongwang Li, Mingyue Ding*

*Corresponding author. Email: dingmy@whu.edu.cn

Published 5 February 2021, *Science* **371**, 610 (2021)
DOI: 10.1126/science.abb3649

This PDF file includes:

Materials and Methods
Figs. S1 to S32
Tables S1 to S6
References

Materials and Methods

Synthesis of Fe₂O₃

The nanorod-shaped Fe₂O₃ was prepared by a hard-template method. Typically, 15 g of Fe(NO₃)₃·9H₂O (AR) was dissolved in 20 mL of water. Subsequently, 4 g of carbon black (cabot vulcan xc-72), which was used as the hard template, was added in the above solution. After stirring at 50 °C for 11 h, the black product was dried in a vacuum oven at 50 °C for 6 h. Finally, the black product was calcined in static air at 550 °C for 5 h, resulting reddish brown powder. The obtained Fe₂O₃ was named as Fe.

Synthesis of Fe@Si

The core-shell Fe₂O₃@SiO₂ was prepared by the modified Stöber method. Typically, 0.5 g of the prepared Fe₂O₃ was dispersed in 300 mL of ethanol (AR) by ultrasonication. Then, 2.5 mL of tetraethoxysilane (TEOS, AR) was added. After stirring under 450 rpm for 4 h, 5 mL of ammonia (25 ~ 28%, AR) and 20 mL of water were added. The mixture was stirred under 450 rpm for another 4 h. Subsequently, the product was washed with ethanol and dried at 100 °C for 11 h. The obtained Fe₂O₃@SiO₂ was named as Fe@Si.

Synthesis of Fe@Si-c

The hydrophobic encapsulation was achieved by further organic modification. The prepared Fe@Si was preheated in a vacuum oven at 150 °C for 11 h. After cooling to 30 °C in the vacuum oven, n-hexane (AR) was added. Subsequently, chlorotrimethylsilane (TMCS, y mL per gram Fe@Si (y = 0.5, 2.0, 6.0)) was added. Then, the mixture was ultrasonic treated (40 kHz, 400W) at room temperature for 3 h. The product was washed with n-hexane and dried in a vacuum oven at 80 °C for 11 h. The obtained catalysts were named as Fe@Si-c_y. The dosage of TMCS was 2.0 mL per gram Fe@Si unless otherwise noted.

Synthesis of FeMn@Si-c

The Mn promoted Fe₂O₃ was prepared by the deposition-precipitation method. Typically, 0.5 g of the prepared Fe₂O₃ was dispersed in 300 mL of water by ultrasonication. Mn(NO₃)₂ (50%, AR) with stoichiometric amount was added. Then, the mixture was placed in an oil bath at 90 °C for 1 h. After adding 2 mL of ammonia, the mixture was aged for 11 h. The product was washed with water and dried at 100 °C for 11 h. The SiO₂-encapsulation and further organic modification were the same as for Fe₂O₃ mentioned above. The obtained catalysts were named as Fe₁Mn_z@Si-c (Mn/Fe molar ratio = z: 1, z = 0.0, 0.1, 0.3, 0.5, 1.0). The Mn/Fe ratio was 0.3 unless otherwise noted.

Catalyst characterization

Textural properties of the catalysts were measured by N₂ physisorption at -196 °C on Micromeritics ASAP 240 instrument. Prior to the measurement, the catalysts were degassed at 300 °C for 10 h under vacuum. X-ray diffraction (XRD) patterns of the powder catalysts were recorded with a PANalytical X'Pert Pro diffractometer over the 2θ range of 10° - 80°. The phases of catalysts were identified based on JCPDS standard cards. X-ray photoelectron spectroscopy (XPS) of the catalysts were recorded on a

Thermal XPS ESCALAB 250Xi Spectrometer. All peaks were corrected by setting the C 1s peak of 284.6 eV as the reference. Mössbauer spectroscopy experiments were carried out in an MR-351 constant-acceleration Mössbauer spectrometer (FAST, Germany).

The test process in Fig. 3D. After “reduction and carburization” or “reduction and carburization and subsequent reaction”, the reactor was cooled to room temperature under N₂ atmosphere. Then, the catalysts were collected and characterized by XRD and Mössbauer spectroscopy. Through such a test-characterization process, we can observe as much as possible the phase composition of catalysts under the actual industrial reaction conditions.

The morphology and elemental composition of the catalysts were characterized by scanning electron microscopy (SEM) on a MIRA3 TESCAN microscope equipped with an Aztec Energy X-Max 20 energy dispersive X-ray spectroscopy (EDS) detector. Transmission electron microscopy (TEM) and high-resolution transmission electron microscopy (HRTEM) images were obtained on a TECNAI G2 F30 with 200 kV accelerating voltage.

In situ diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) measurements were accomplished in an in situ sample cell on a Thermo IS10 spectrometer. 10 mg powder catalyst was placed into the sample cell. Each measurement was performed after heating the sample cell in situ up to 400 °C under a flow of He (30 mL/min) with a ramp of 10 °C/min and keeping it at 400 °C for 2 h. The spectrum was then recorded at room temperature. Fourier transform infrared spectra (FTIR) was performed on a NICOLET 5700 FTIR Spectrometer. Prior to the measurement, the sample was mixed with potassium bromide and pressed into a wafer.

Water-droplet contact angle tests were performed on the JY-82B contact angle goniometer. An Intrinsic DVS instrument from Surface Measurement Systems was used to perform the water vapor adsorption and desorption experiments over a humidity range of 0 ~ 95%. Each humidity step was made if less than a 0.002% weight change occurred over 10 minutes, with a maximum hold time of 240 minutes.

Thermogravimetric (TG) curves were obtained on a METTLER TGA/DSC in flowing air with a heating rate of 10 °C/min. Raman spectra of the catalysts were performed on a Thermo DXR.

Catalytic tests

Fischer-Tropsch synthesis (FTS) reactions were performed in a stainless fixed-bed reactor (inner diameter = 10 mm, length = 40 cm). Typically, 0.5 g catalyst (40-60 meshes) diluted with 0.5 g quartz sand was used. Prior to reaction, the catalyst was reduced in situ at 350 °C for 10 h in a pure H₂ flow at atmospheric pressure. After reduction, the reactor was cooled to 200 °C. Then the syngas H₂/CO/N₂ (containing 5 vol% N₂ as the internal standard) was fed into the reactor. The reaction products were detected by two gas chromatographs (FULI GC 97). One gas chromatograph was equipped with a thermal conductivity detector (TCD) and a flame ionization detector (FID). Porapak Q and 5A MolSieve packed column were connected to TCD, while RB-PLOT Al₂O₃ capillary column was connected to FID. The other gas chromatograph was equipped with a RB-5 capillary column and a FID. After reaction, the spent catalyst was transformed into a small glass bottle and then inflated with N₂. The bottle was put into a Teflon flask deoxygenated by Fe powder and then vacuumed and filled with N₂. The

carbon balance calculated in each test was between 96% and 98%. The CO conversion and product selectivity were calculated on a carbon-atom basis. The selectivity to oxygenates was below 2% C and has been excluded from the reported product selectivities.

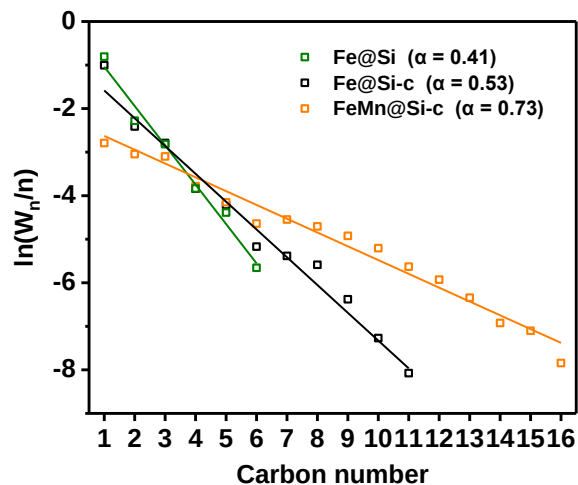


Fig. S1.

The ASF plots and relevant chain growth probability over Fe@Si, Fe@Si-c and FeMn@Si-c (corresponding to Fig. 1B). Reaction conditions: 320 °C, 2.0 MPa, 4000 ml h⁻¹ g⁻¹, H₂/CO = 2.

Note: As shown in Fig. S1, Fe@Si-c exhibited a higher chain growth probability ($\alpha = 0.53$) than Fe@Si ($\alpha = 0.41$). Due to the inhibition of WGS reaction via hydrophobic treatment on Fe@Si, the H₂ amount produced by the WGS reaction decreased, suppressing the hydrogenation and chain termination reactions, thereby enhancing the chain growth probability. The addition of Mn promoter further increased the α value to 0.73 over FeMn@Si-c. Besides, the amount of CH₄ obtained over FeMn@Si-c was lower obviously than that over Fe@Si and Fe@Si-c. All of these results suggested that the hydrophobic treatment and the addition of Mn promoter effectively increased the chain growth probability and suppressed the formation of CH₄.

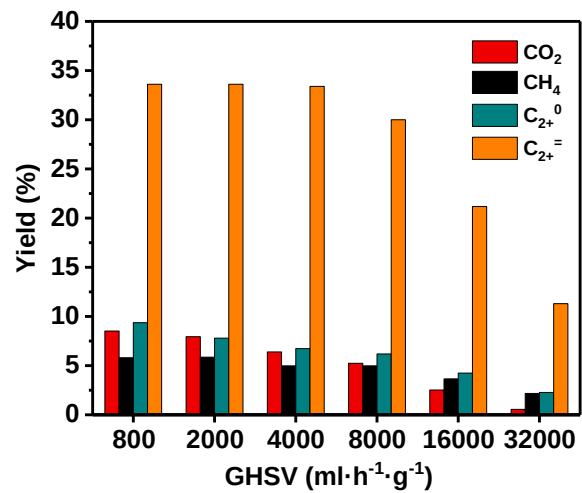


Fig. S2.

Effect of gas hourly space velocity (GHSV) on the yield of products. Reaction conditions: 320 °C, 2.0 MPa, H₂/CO = 2.

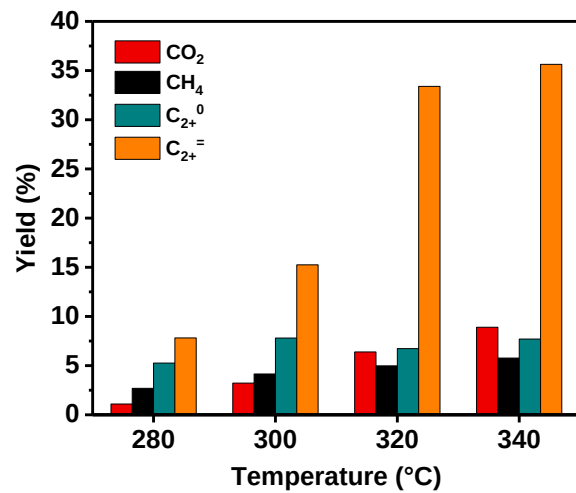


Fig. S3.

Effect of reaction temperature on the yield of products. Reaction conditions: 2.0 MPa, 4000 ml h⁻¹ g⁻¹, H₂/CO = 2.

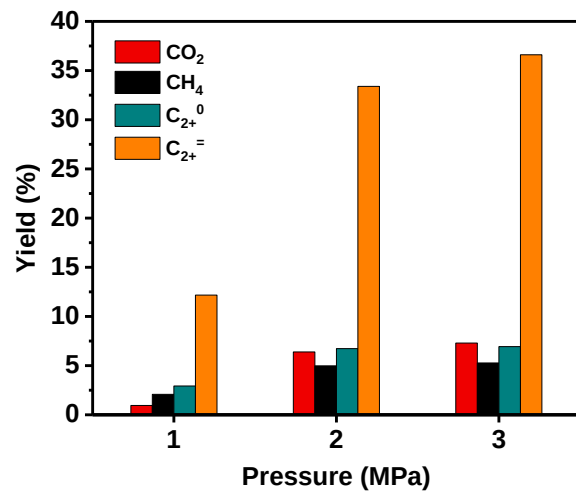


Fig. S4.

Effect of reaction pressure on the yield of products. Reaction conditions: 320 °C, 4000 ml h⁻¹ g⁻¹, H₂/CO = 2.

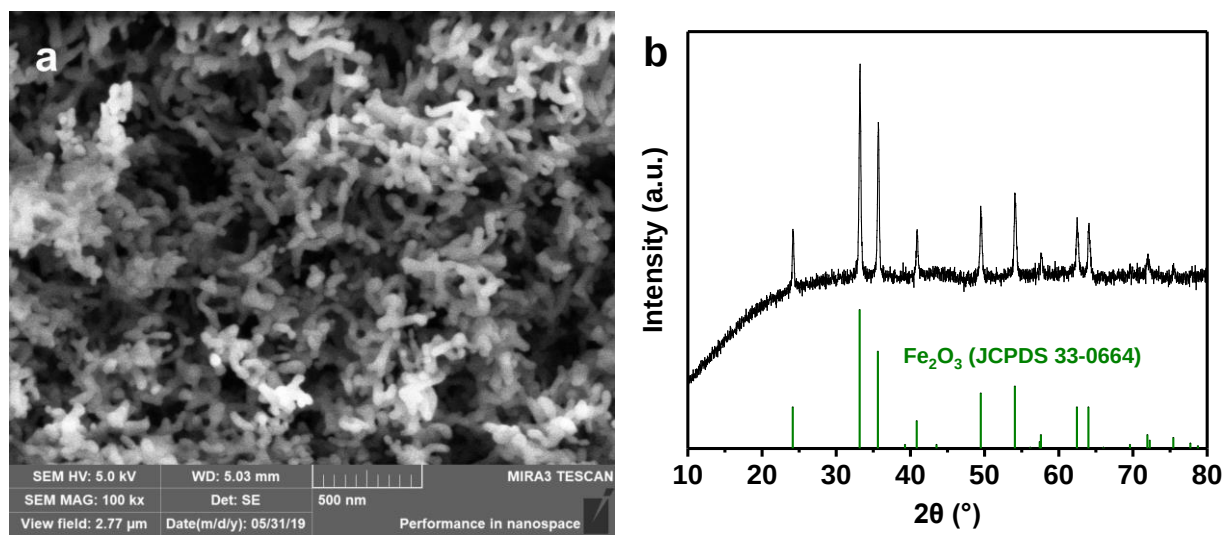


Fig. S5.

(a) SEM image and (b) XRD pattern of the Fe₂O₃ nanorods.

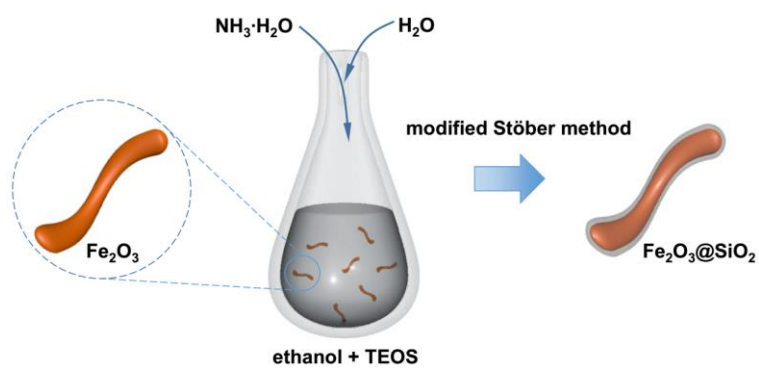


Fig. S6.
Procedures for the synthesis of core-shell $\text{Fe}_2\text{O}_3@\text{SiO}_2$.

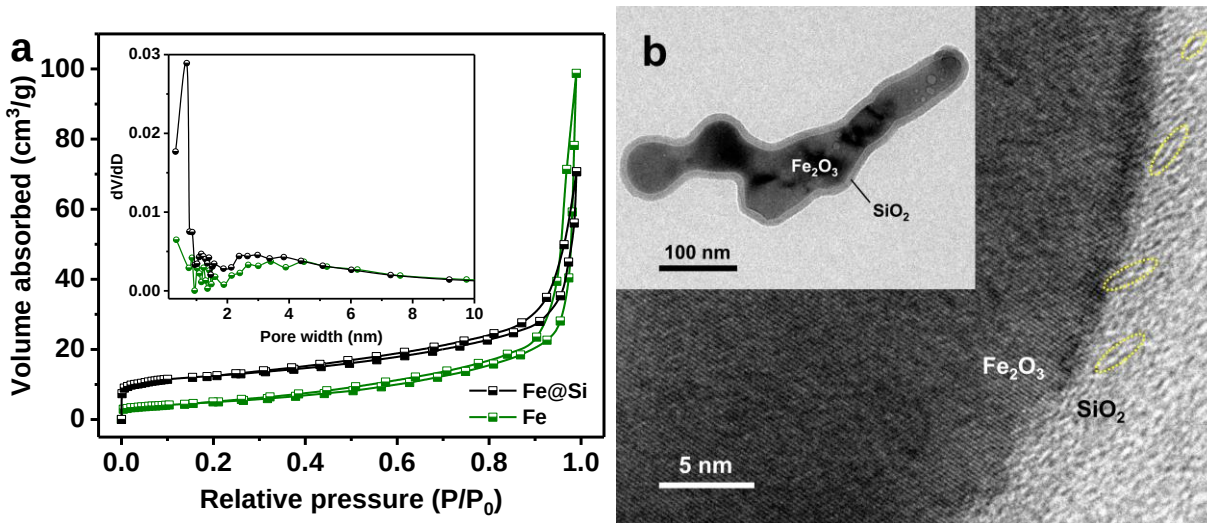


Fig. S7.

(a) N $_2$ adsorption-desorption isotherms and pore size distribution of Fe and Fe@Si. (b) TEM and HRTEM images of Fe@Si.

Note: As shown in Fig. S7a, after the encapsulation of SiO $_2$ on Fe $_2\text{O}_3$, more micropores with pore size ranging from 0.3 to 2 nm appeared in the pore size distribution curve, suggesting that micropores were presented in the SiO $_2$ shell. The HRTEM image also confirmed that many narrow micropores (marked by the golden circle in Fig. S7b) were distributed in the amorphous SiO $_2$ shell.

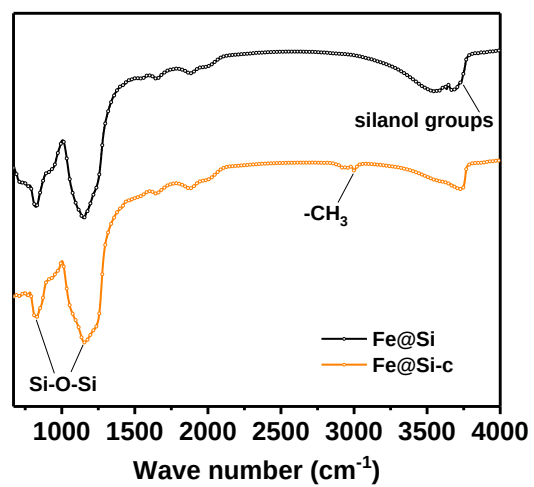


Fig. S8.
DRIFTS of Fe@Si and Fe@Si-c.

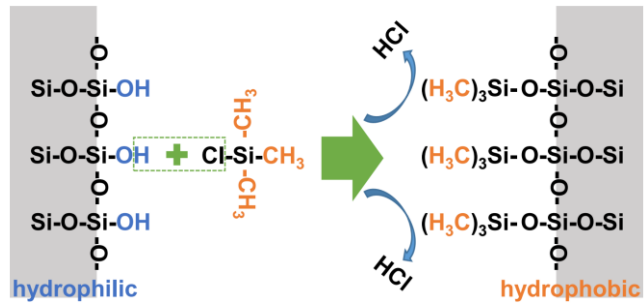


Fig. S9.

Hydrophobic modification through silanization reaction.

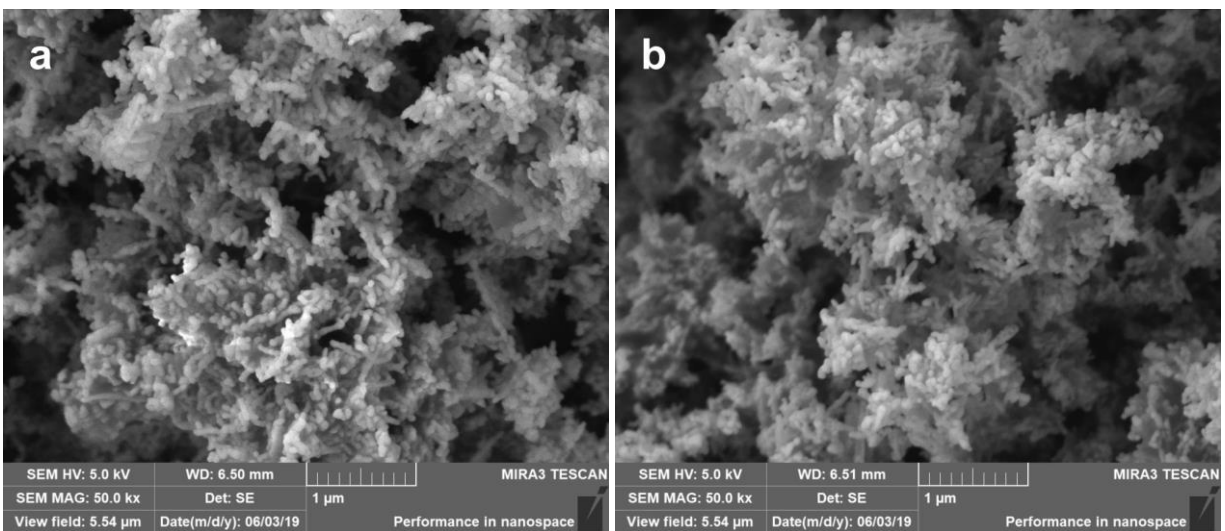


Fig. S10.

SEM images of (a) Fe@Si and (b) Fe@Si-c.

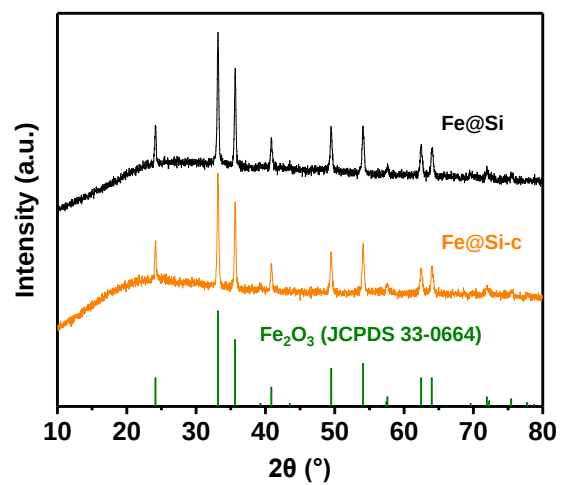


Fig. S11.
XRD patterns of Fe@Si and Fe@Si-c.

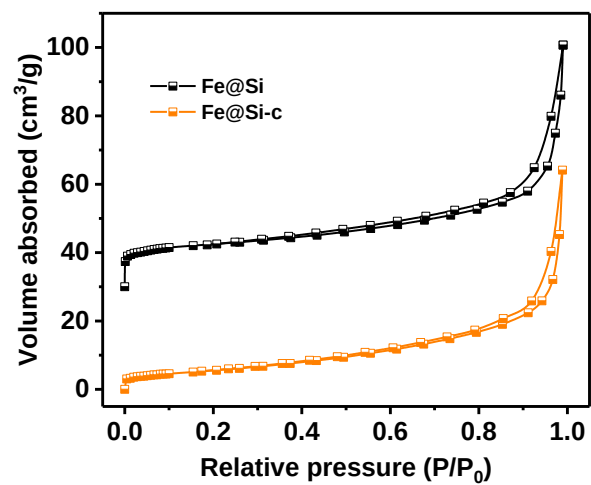


Fig. S12.

N₂ adsorption-desorption isotherms of Fe@Si and Fe@Si-c.

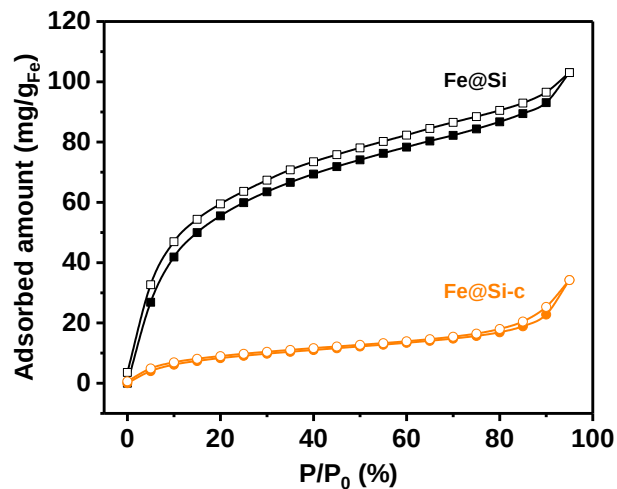


Fig. S13.

The water vapor adsorption and desorption tests on Fe@Si and Fe@Si-c. The solid and hollow markings show the adsorption and desorption data, respectively.

Note: As shown in Fig. S13, after the hydrophobic modification, the water vapor adsorption capacity of Fe@Si-c dramatically decreased compared with the hydrophilic Fe@Si. At the humidity less than 30%, the adsorbed amount of water vapor over Fe@Si was about 6.5 times that over Fe@Si-c, suggesting that the hydrophobic modification could suppress obviously the adsorption of water molecules. These results were consistent with the previous reports (35-38) that the introduced nonpolar -CH₃ groups could hinder the adsorption of polar water molecules.

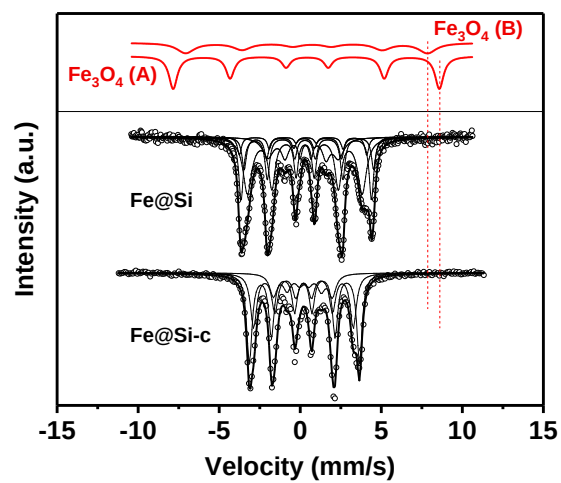


Fig. S14.

Mössbauer spectra of the catalysts after reduction and carburization (corresponding to Fig. 3D).

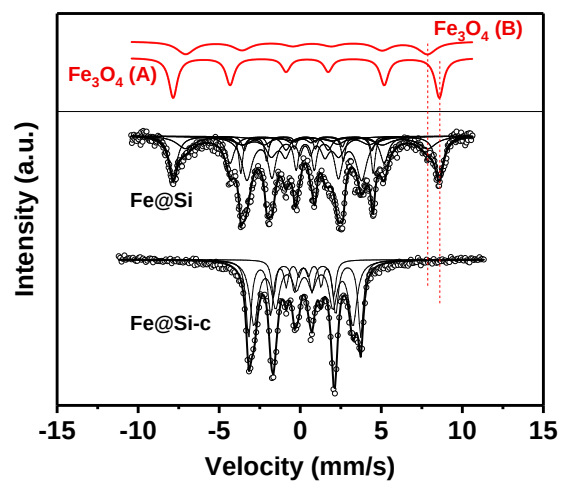


Fig. S15.

Mössbauer spectra of the catalysts after reduction and carburization and subsequent reaction (corresponding to Fig. 3D).

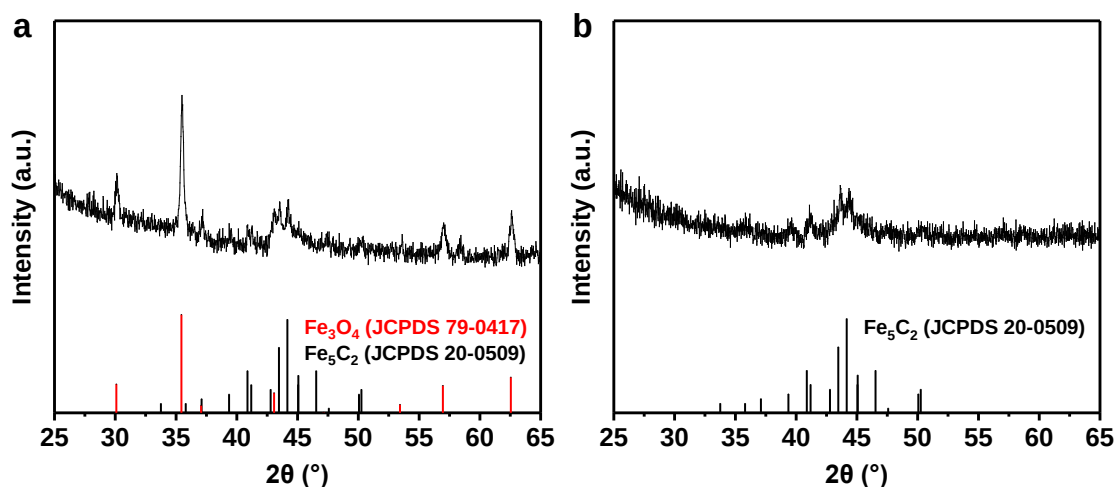


Fig. S16.

XRD patterns of (a) Fe@Si and (b) Fe@Si-c after the test in Fig. 4A. Reaction conditions: 320 °C, 2.0 MPa, H₂/CO = 1.

Note: In Fig. 4A, the CO₂ selectivity-CO conversion curve was obtained by adjusting the flow rate at a constant H₂/CO ratio. Generally, the traditional Fe-based catalysts were converted into a mixture of iron carbides and oxides during reaction due to the carburization by CO and oxidation by H₂O (13, 24, 25, 27). The carbides oxidation could be easily achieved in the initial stage after exposure of syngas under conventional reaction conditions. In order to obtain conveniently the CO₂ selectivity-CO conversion curve, we checked firstly the relationship between the CO₂ selectivity and CO conversion at conventional reaction conditions (usually at a higher CO conversion), and then analyzed the CO₂ selectivity-CO conversion curve by adjusting the flow rate at a constant H₂/CO ratio. Thus, the data of Fe@Si at CO conversion of ~5% was collected continuously after collecting the data at higher CO conversion levels by adjusting the flow rate. Due to the coexistence of iron carbides and Fe₃O₄ over the Fe@Si catalyst at a higher CO conversion, both iron carbides and Fe₃O₄ still coexisted when collecting the data at CO conversion of ~5% by adjusting the flow rate (as shown in Fig. S16a). Thus, the produced secondary CO₂ was existed over the Fe@Si catalyst. Besides, a primary CO₂ selectivity of about 10% was observed at very low CO conversion over the Fe@Si catalyst, which might be attributed to the presence of Fe₃O₄ phase, as shown in Fig. 4A. Because the hydrophobic shell coated on Fe@Si-c could restrain the oxidation of iron carbides to Fe₃O₄, it was easy to analyze the effect of iron carbides on the CO₂ selectivity. Fe@Si-c was only composed of iron carbides after such test process (as shown in Fig. S16b). No formation of CO₂ was observed at a lower CO conversion (less than 5%), suggesting that only iron carbides existed in the Fe@Si-c catalyst are free of primary CO₂.

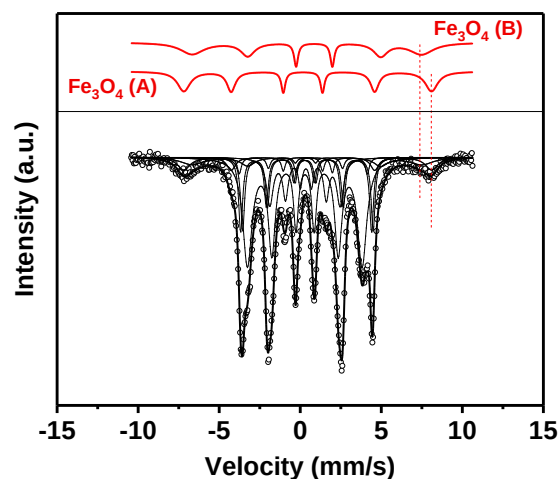


Fig. S17.

Mössbauer spectrum of Fe@Si-c at CO conversion of about 78%. Reaction conditions: 320 °C, 2.0 MPa, H₂/CO = 1.

Note: At relatively high CO conversion level (about 78%), both iron carbides (85.2%) and little Fe₃O₄ (14.8%) were detected in the Mössbauer spectrum of the spent Fe@Si-c catalyst (Fig. S17 and Table S5), suggesting that oxidation of iron carbides began to occur at much higher CO conversion. At a higher CO conversion level, the amount of water formed would be high, which resulted inevitably in partial oxidation of iron carbides into magnetite before leaving the hydrophobic layer. The formation of Fe₃O₄ enhanced the WGS reaction and promoted the formation of by-produced CO₂. As a result, the CO₂ selectivity reached over 40% at CO conversion of about 78% (Fig. S18).

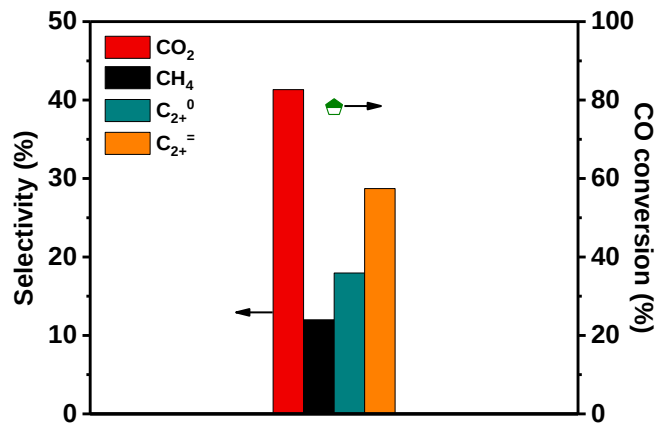


Fig. S18.

Product distribution of Fe@Si-c at relatively high CO conversion. Reaction conditions: 320 °C, 2.0 MPa, H₂/CO = 1.

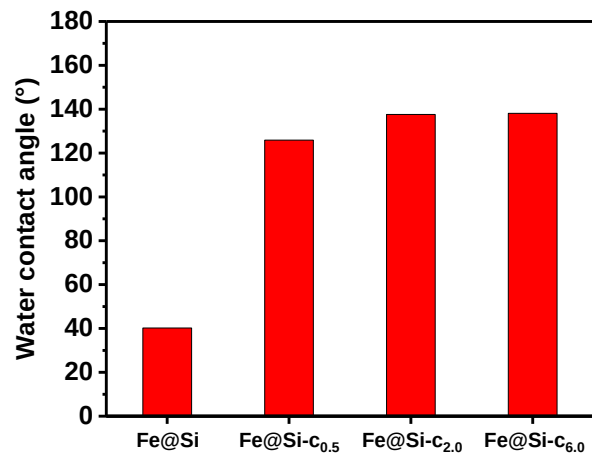


Fig. S19.

Water contact angle over Fe@Si-c with different exposures of TMCS.

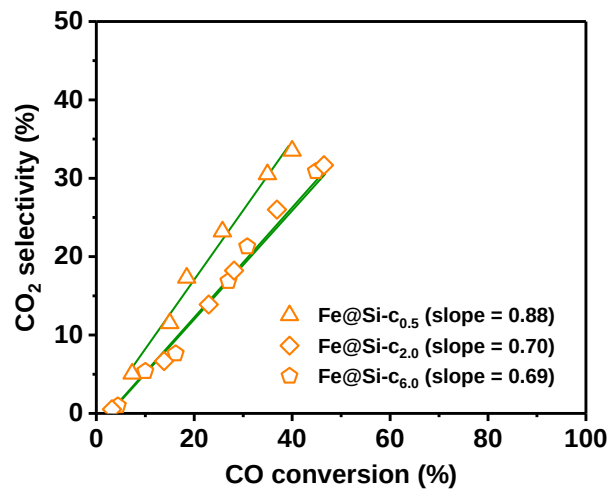


Fig. S20.

CO₂ selectivity-CO conversion curve over Fe@Si-c with different exposures of TMCS.
Reaction conditions: 320 °C, 2.0 MPa, H₂/CO = 1.

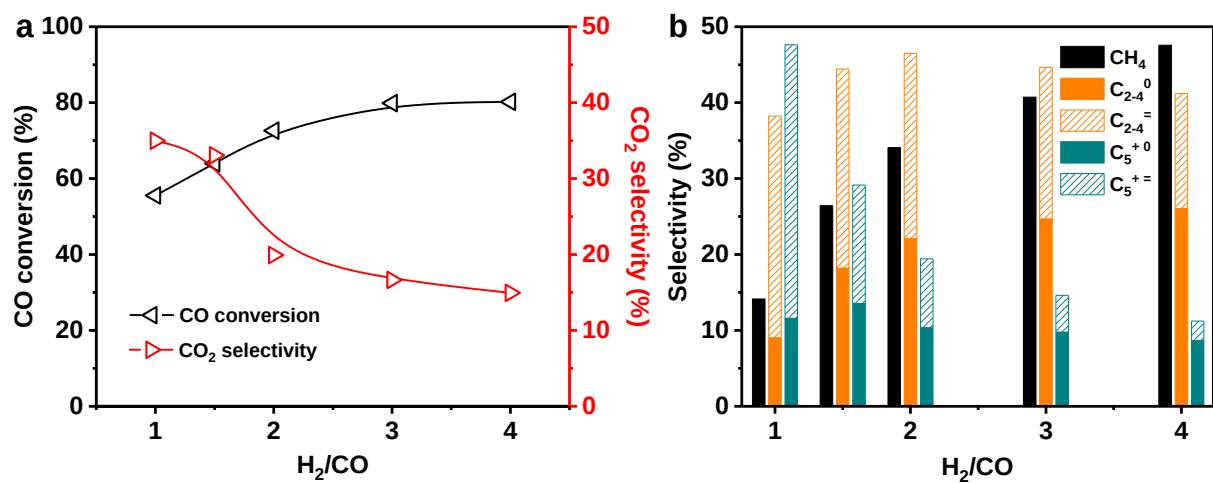


Fig. S21.

Effect of H_2/CO ratio in syngas on (a) CO conversion and CO_2 selectivity and (b) hydrocarbons distribution of Fe@Si-c. Reaction conditions: 320 °C, 2.0 MPa, 4000 ml $h^{-1} g^{-1}$.

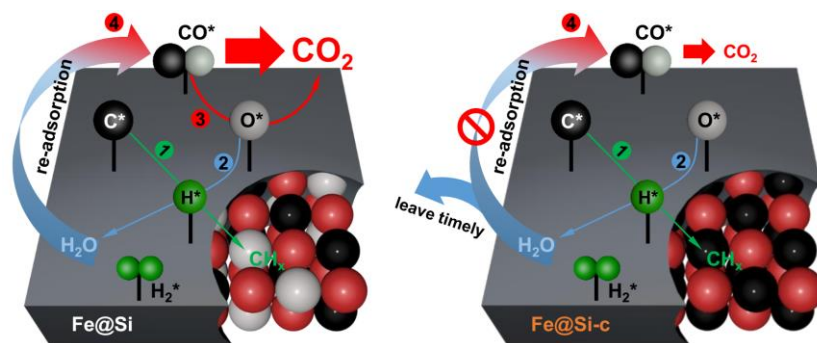


Fig. S22.

Plausible mechanism of hydrophobic modification inhibiting the CO₂ formation.

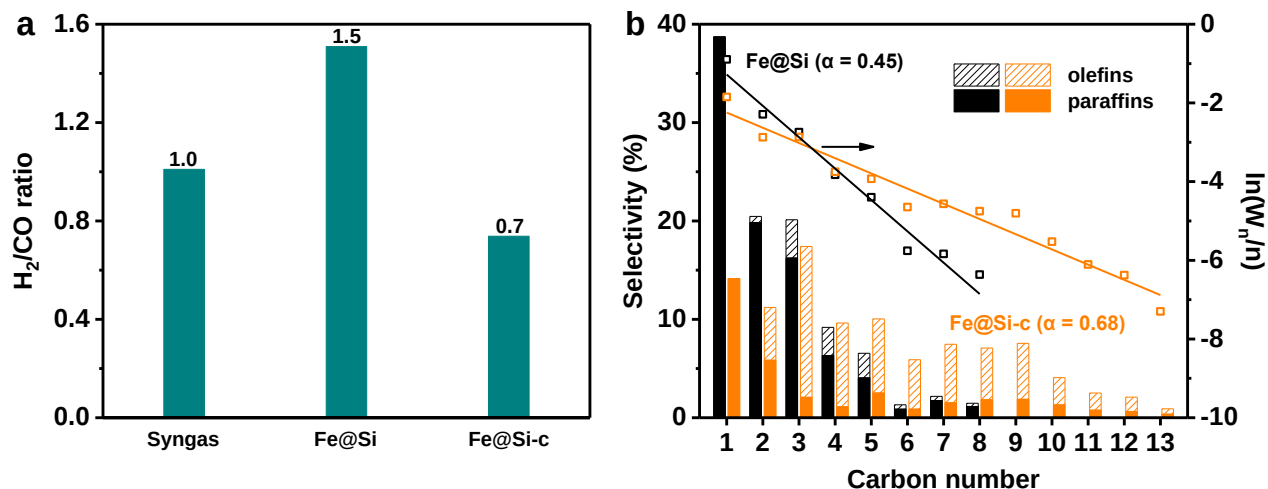


Fig. S23.

Comparison of (a) H_2/CO ratio at the gas outlet and (b) hydrocarbons distribution obtained over Fe@Si and Fe@Si-c. Reaction conditions: 320 °C, 2.0 MPa, $H_2/CO = 1$.

Note: The H_2/CO ratio at the gas outlet in Fig. S23a referred to the mole ratio of H_2 and CO at the gas outlet. And the amount of H_2 and CO at the outlet were determined online by the gas chromatograph. As shown in Fig. S23b, Fe@Si-c exhibited higher chain growth probability and olefins selectivity than Fe@Si. Due to the inhibition of WGS reaction via hydrophobic treatment on Fe@Si, the H_2 amount produced by the WGS reaction decreased, suppressing the hydrogenation and chain termination reactions, thereby enhancing the chain growth probability and olefins production.

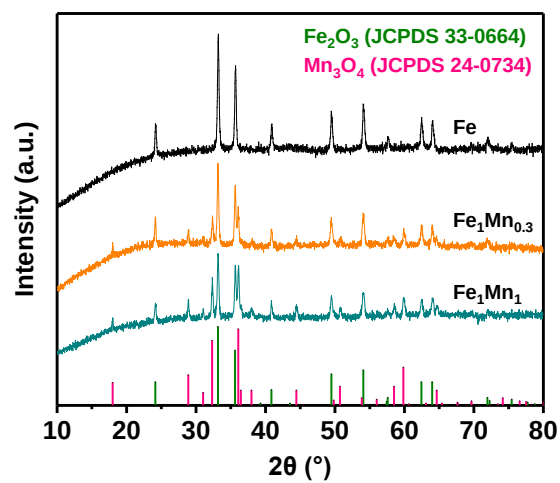


Fig. S24.

XRD patterns of the Fe₁Mn_z catalysts with different Mn/Fe ratios.

Note: As shown in Fig. S24, the added Mn was presented in the form of Mn₃O₄ phase, and no diffraction peak related to Fe-Mn solid solution was detected.

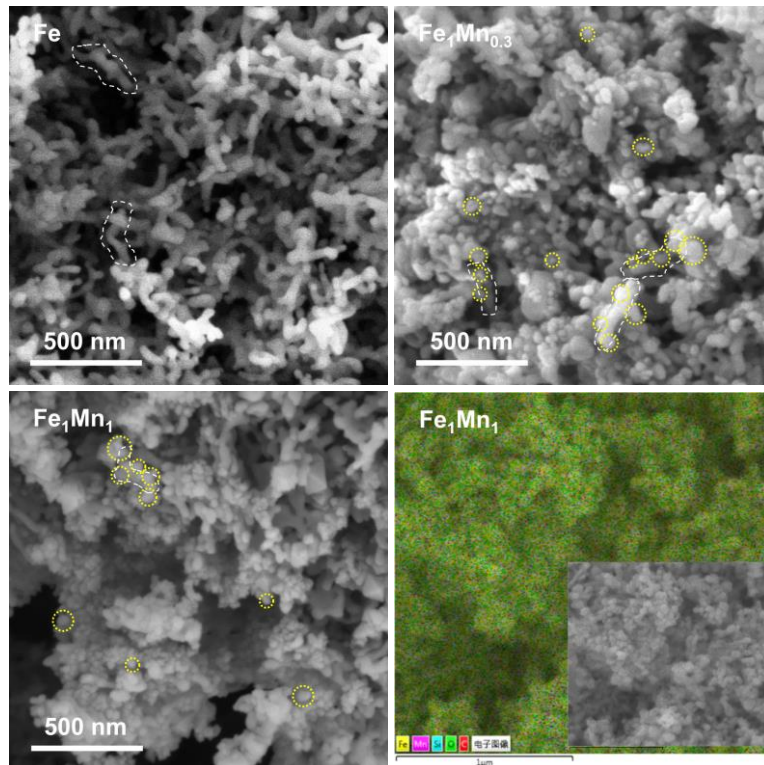


Fig. S25.

SEM images of the Fe_1Mn_z catalysts with different Mn/Fe ratios and relevant EDS elemental mapping. Scale bar in EDS: 1 μm .

Note: As shown in Fig. S25, the single Fe_2O_3 was presented in the form of nanorod-shape (marked by the white dotted line). After the addition of Mn promoter, many microspheres (marked by the golden circle) homogeneously located in the voids between the aggregated Fe_2O_3 nanorods, suggesting that the added Mn was presented in the form of microspheres.

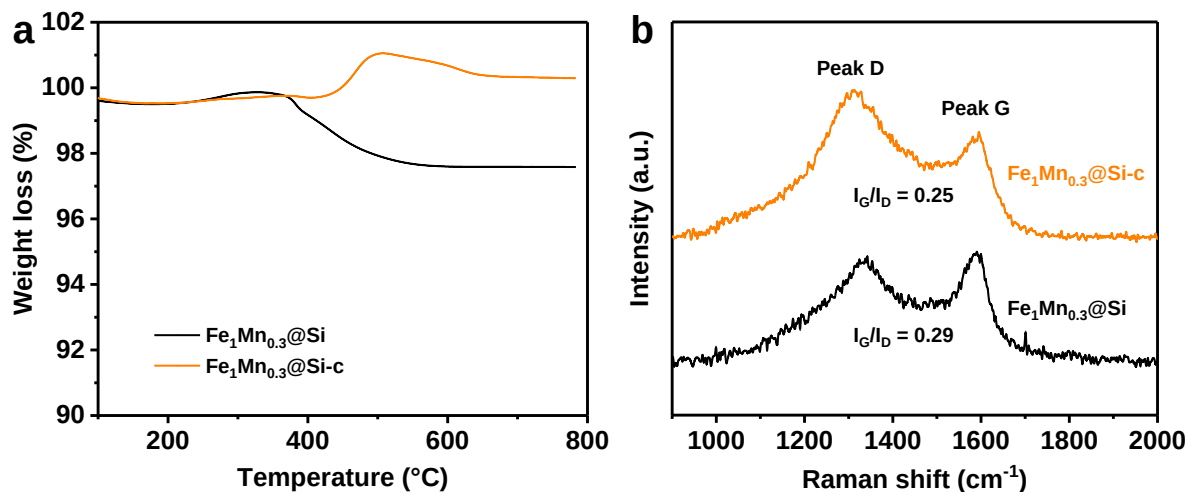


Fig. S26.

(a) TG curves and (b) Raman spectra of the spent $\text{Fe}_1\text{Mn}_{0.3}\text{@Si}$ and $\text{Fe}_1\text{Mn}_{0.3}\text{@Si-c}$.

Note: As shown in Fig. S26a, the coke contents of the two spent catalysts were less than 2.3%, implying that little coke was deposited on the two catalysts. The weight-gain region of the spent $\text{Fe}_1\text{Mn}_{0.3}\text{@Si-c}$ might be attributed to the oxidation process of iron carbides to iron oxides in air. As shown in Fig. S26b, two peaks appeared at 1330 cm^{-1} and 1600 cm^{-1} , which were attributed to the disordered (D type) and ordered (G type) carbonaceous species, respectively (39). $\text{Fe}_1\text{Mn}_{0.3}\text{@Si-c}$ exhibited a lower I_G/I_D ratio than $\text{Fe}_1\text{Mn}_{0.3}\text{@Si}$, indicating more iron carbides active sites. Besides, the carbon balance calculated in each test was between 96% and 98%. As discussed above, the hydrophobic modification could decrease the water content in the vicinity of catalysts and reduce the H_2 amount produced by the WGS reaction. TG and Raman spectra results suggested that the concentration reduction of H_2O and H_2 on the hydrophobic catalysts did not affect obviously the coke deposition and carbon balance during reaction.

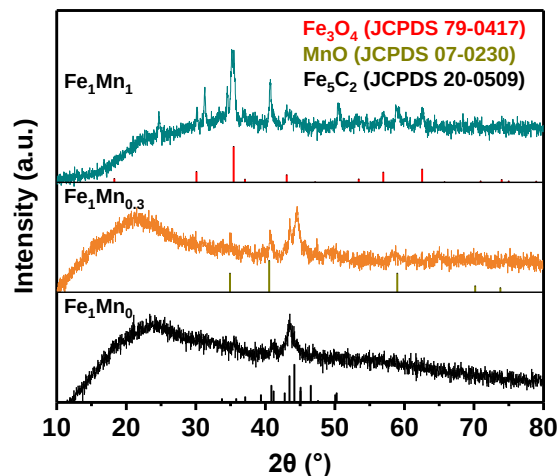


Fig. S27.

XRD patterns of the spent $\text{Fe}_1\text{Mn}_z@\text{Si-c}$ with different Mn/Fe ratios. Reaction conditions: 320 °C, 2.0 MPa, $\text{H}_2/\text{CO} = 2$.

Note: As shown in Fig. S27, only iron carbides were detected in the XRD patterns of the spent $\text{Fe}_1\text{Mn}_z@\text{Si-c}$ catalysts with Mn/Fe ratio ≤ 0.3 . However, both Fe_3O_4 and iron carbides were detected in the XRD pattern of the spent $\text{Fe}_1\text{Mn}_1@\text{Si-c}$ catalyst. This was attributed to that moderate addition of Mn (Mn/Fe ratio ≤ 0.3) did not influence the homogeneous encapsulation of SiO_2 shell, while excessive Mn promoter hindered the effective contact between the hydrophobic shell and iron species core (Fig. S29). The insufficient shell on the highly promoted catalysts could only inhibit partially the adsorption of water, resulting in partial oxidation of iron carbides into Fe_3O_4 .

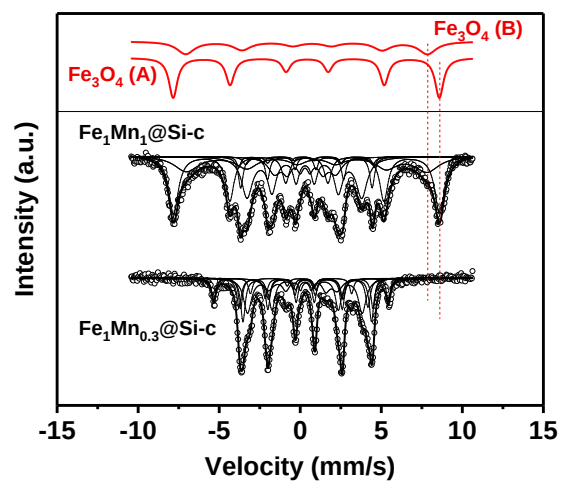


Fig. S28.

Mössbauer spectra of the spent $\text{Fe}_1\text{Mn}_z@\text{Si-c}$ with different Mn/Fe ratios. Reaction conditions: 320 °C, 2.0 MPa, $\text{H}_2/\text{CO} = 2$.

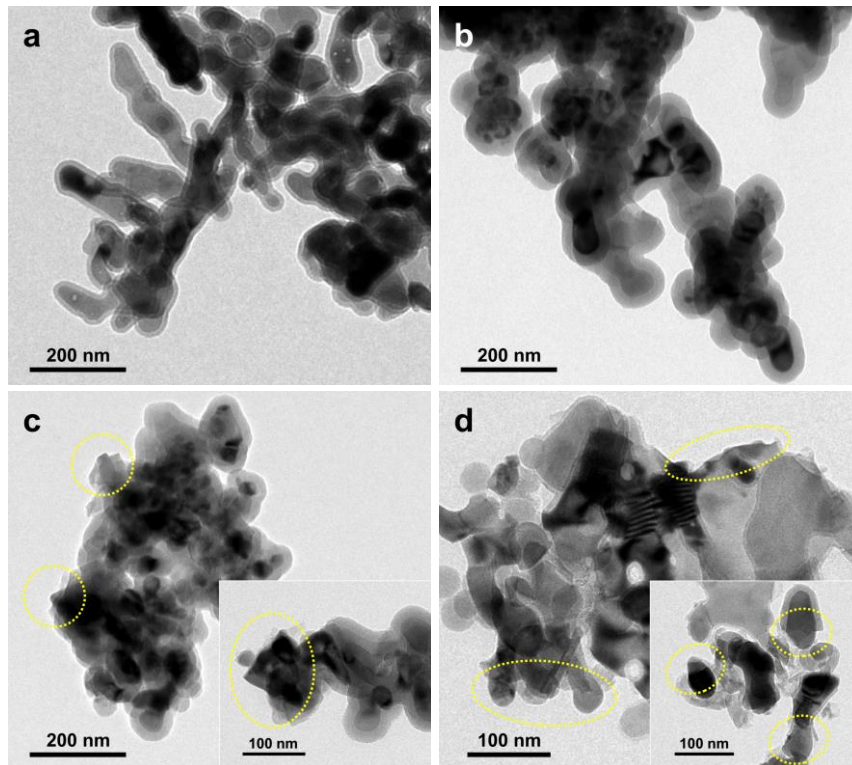


Fig. S29.

TEM images of (a) Fe@Si-c, (b) Fe₁Mn_{0.3}@Si-c, (c) Fe₁Mn_{0.5}@Si-c and (d) Fe₁Mn₁@Si-c.

Note: As shown in Fig. S29, a and b, moderate addition of Mn (Mn/Fe ratio ≤ 0.3) did not influence the homogeneous encapsulation of SiO₂ shell. However, excessive addition of Mn resulted easily in the aggregation of nanorods into large particles (Fig. S29, c and d). Besides, no sufficient shell formation was achieved on the highly promoted catalysts (marked by the golden circle), suggesting that excessive Mn promoter hindered the effective contact between the hydrophobic shell and iron species core.

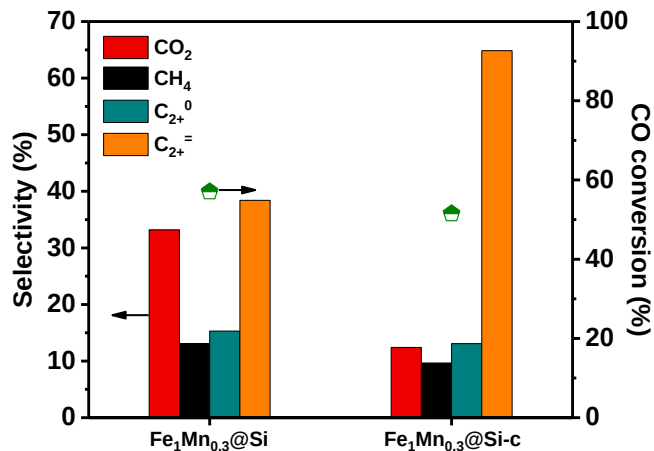


Fig. S30.

Comparison of the product distribution (including CO₂) obtained over Fe₁Mn_{0.3}@Si and Fe₁Mn_{0.3}@Si-c. Reaction conditions: 320 °C, 2.0 MPa, H₂/CO = 2.

Note: As shown in Fig. S30, the CO₂ selectivity of Fe₁Mn_{0.3}@Si-c was obviously lower than that of Fe₁Mn_{0.3}@Si at a similar CO conversion level, suggesting that the WGS activity was significantly inhibited via the hydrophobic treatment. In addition, compared with Fe₁Mn_{0.3}@Si, Fe₁Mn_{0.3}@Si-c exhibited a suppressed CH₄ selectivity and enhanced olefins selectivity, which was a result of that the H₂ amount produced by the WGS reaction decreased, thereby suppressing the overhydrogenation and chain termination reactions. These results were consistent with the comparison results of Fe@Si and Fe@Si-c mentioned above.

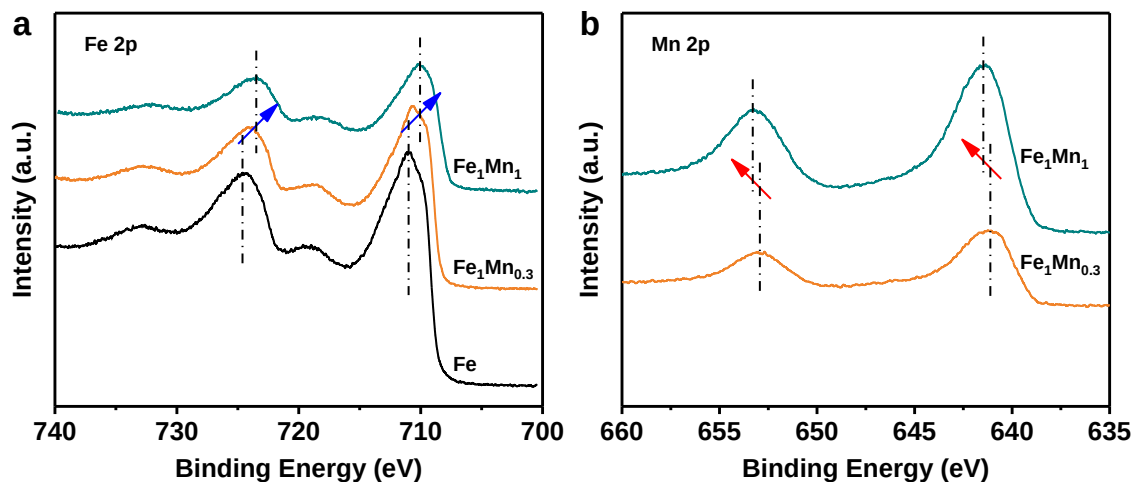


Fig. S31.

XPS spectra of (a) Fe 2p and (b) Mn 2p on the Fe₁Mn_z catalysts with different Mn/Fe ratios.

Note: As shown in Fig. S31, as the increase of Mn content, the Fe 2p peaks gradually shifted to a lower binding energy, while the Mn 2p peaks shifted toward a higher binding energy. These results were attributed to the electron transfer from Mn to Fe atoms (13, 14). This process facilitated the dissociated adsorption of CO and weakened hydrogenation, in favor of C-C coupling reaction and olefins production (14).

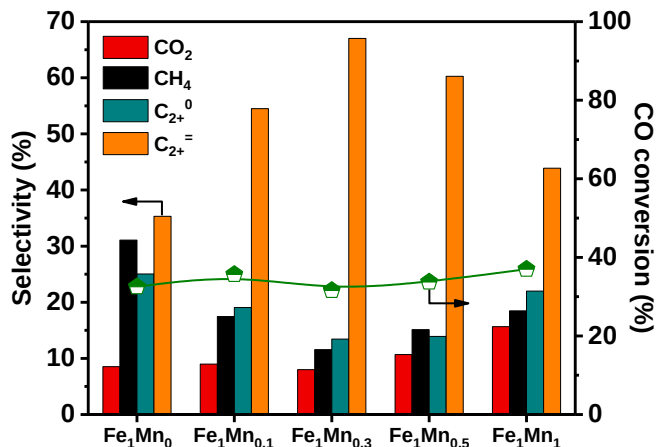


Fig. S32.

Catalytic performance of Fe₁Mn_z@Si-c with different Mn/Fe ratios. Reaction conditions: 320 °C, 2.0 MPa, H₂/CO = 2.

Note: As shown in Fig. S32, under similar CO conversion level, moderate addition of Mn (Mn/Fe ratio ≤ 0.3) had little influence on the formation of CO₂. Whereas, excessive addition of Mn resulted in the increase of CO₂ selectivity from ~8 to ~16%. This was attributed to that moderate addition of Mn (Mn/Fe ratio ≤ 0.3) did not influence the homogeneous encapsulation of SiO₂ shell, while excessive Mn promoter hindered the effective contact between the hydrophobic shell and iron species core (Fig. S29). The insufficient shell on the highly promoted catalysts could only inhibit partially the adsorption of water, resulting in partial oxidation of iron carbides into Fe₃O₄ (Figs. S27 and S28) and the enhancement of WGS reaction. Thus, excessive addition of Mn resulted in the increase of CO₂ selectivity.

More importantly, moderate addition of Mn promoted the formation of olefins and suppressed that of CH₄. This was attributed to the enhanced electron transfer from Mn to Fe atoms (Fig. S31), which facilitated the dissociated adsorption of CO and weakened the hydrogenation ability, favoring the C-C coupling reaction and olefins production (14). While, excessive addition of Mn resulted in the enhancement of WGS reaction, which increased the H₂ amount produced by the WGS reaction, thereby facilitating the hydrogenation reaction and suppressing the olefins production.

Table S1.

Comparison of catalytic performance with the previous works.

Catalyst	T (°C)	P (MPa)	GHSV (h ⁻¹)	CO conv. (%)	CO ₂ sel. (%)	Yield of hydrocarbons (%)	Olefins sel. ^d (%)	Yield of olefins (%)	Ref.
ZnCrO _x /SAPO	400	2.5	6828	17	~45	~9.35	80 ^a	~7.48 ^a	(5)
MnO _x /SAPO	400	2.5	4800	8.5	41	5.02	79.2 ^a	3.98 ^a	(6)
ZnZrO _x /SAPO	400	1.0	3600	9.5	45	5.23	63 ^a	3.29 ^a	(7)
CoMn	250	0.1	2000	31.8	47.3	16.8	60.8 ^a	10.2 ^a	(8)
Fe/ α -Al ₂ O ₃	340	2.0	1500	80	40	48.0	53 ^a	25.4 ^a	(9)
Fe-K/NCNTs	300	0.1	4200	16.5	23.6	12.6	54.6 ^a	6.88 ^a	(10)
Mn/Fe ₃ O ₄	320	1.0	4480	41.5	37.8	25.8	60.1 ^a	15.5 ^a	(13)
Fe ₃ O ₄ @MnO ₂	280	2.0	3000	67.9	47.1	35.9	79.3 ^b	28.5 ^b	(14)
KFe@C	340	2.0	30000	~82	47.4	~43.1	44.6 ^a	~19.2 ^a	(15)
Fe/hNCNC	350	0.1	12000	3.5	39.4	2.12	54.1 ^a	1.15 ^a	(16)
FeMn@Si-c	320	3.0	4000	56.1	13.0	48.8	75 ^b	36.6 ^b	TW ^c

^a Light olefins; ^b Total olefins; ^c This work; ^d Olefins selectivity in hydrocarbons.

Table S2.

Detailed Mössbauer parameters (corresponding to Fig. 3, B and C).

Catalyst	Assignment	Mössbauer parameters			Spectral contribution (%)
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	
Fe@Si	Fe ₃ O ₄ (A)	0.39	0.02	508	33.8
	Fe ₃ O ₄ (B)	0.70	-0.46	469	26.9
	χ -Fe ₅ C ₂ (I)	0.34	0.14	254	5.4
	χ -Fe ₅ C ₂ (II)	0.30	-0.15	220	22.6
	χ -Fe ₅ C ₂ (III)	0.21	-0.16	122	7.7
	θ -Fe ₃ C (a)	0.20	0.30	260	3.6
Fe@Si-c	χ -Fe ₅ C ₂ (I)	0.40	0.16	250	18.6
	χ -Fe ₅ C ₂ (II)	0.29	-0.01	220	52.5
	χ -Fe ₅ C ₂ (III)	0.30	-0.08	136	14.7
	θ -Fe ₃ C (a)	0.24	0.20	256	5.7
	θ -Fe ₃ C (b)	0.29	0.05	249	8.5

Table S3.

Detailed Mössbauer parameters (corresponding to Fig. S14).

Catalyst	Assignment	Mössbauer parameters			Spectral contribution (%)
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	
Fe@Si	χ -Fe ₅ C ₂ (I)	0.33	0.07	250	23.8
	χ -Fe ₅ C ₂ (II)	0.29	-0.02	220	40.8
	χ -Fe ₅ C ₂ (III)	0.25	-0.15	135	17.7
	θ -Fe ₃ C (a)	0.38	0.30	248	11.9
	θ -Fe ₃ C (b)	0.27	-0.19	244	5.8
Fe@Si-c	χ -Fe ₅ C ₂ (I)	0.22	-0.07	210	48.8
	χ -Fe ₅ C ₂ (II)	0.20	0.02	190	34.3
	χ -Fe ₅ C ₂ (III)	0.20	0.08	113	16.9

Table S4.

Detailed Mössbauer parameters (corresponding to Fig. S15).

Catalyst	Assignment	Mössbauer parameters			Spectral contribution (%)
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	
Fe@Si	Fe ₃ O ₄ (A)	0.40	-0.06	510	21.8
	Fe ₃ O ₄ (B)	0.56	-0.34	464	14.1
	χ -Fe ₅ C ₂ (I)	0.35	0.13	254	9.4
	χ -Fe ₅ C ₂ (II)	0.28	-0.06	220	28.9
	χ -Fe ₅ C ₂ (III)	0.30	-0.01	130	13.5
	θ -Fe ₃ C (a)	0.40	0.17	245	4.3
	θ -Fe ₃ C (b)	0.20	0.30	243	8.0
Fe@Si-c	χ -Fe ₅ C ₂ (I)	0.23	-0.09	214	37.7
	χ -Fe ₅ C ₂ (II)	0.22	0.06	189	47.8
	χ -Fe ₅ C ₂ (III)	0.21	0.00	115	14.5

Table S5.

Detailed Mössbauer parameters (corresponding to Fig. S17).

Catalyst	Assignment	Mössbauer parameters			Spectral contribution (%)
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	
Fe@Si-c	Fe ₃ O ₄ (A)	0.31	0.29	475	7.0
	Fe ₃ O ₄ (B)	0.64	-0.45	440	7.8
	χ -Fe ₅ C ₂ (I)	0.34	0.04	251	12.8
	χ -Fe ₅ C ₂ (II)	0.29	-0.02	220	43.1
	χ -Fe ₅ C ₂ (III)	0.32	-0.06	135	13.9
	θ -Fe ₃ C (a)	0.35	0.24	250	12.6
	θ -Fe ₃ C (b)	0.25	-0.13	245	2.8

Table S6.

Detailed Mössbauer parameters (corresponding to Fig. S28).

Catalyst	Assignment	Mössbauer parameters			Spectral contribution (%)
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	
Fe ₁ Mn _{0.3} @Si-c	χ -Fe ₅ C ₂ (I)	0.30	0.19	254	12.9
	χ -Fe ₅ C ₂ (II)	0.30	-0.02	220	34.0
	χ -Fe ₅ C ₂ (III)	0.25	-0.20	132	12.9
	θ -Fe ₃ C (a)	0.36	0.13	246	21.0
	θ -Fe ₃ C (b)	0.27	-0.13	247	14.4
	α -Fe	0.05	/	335	4.8
Fe ₁ Mn ₁ @Si-c	Fe ₃ O ₄ (A)	0.40	-0.02	510	31.3
	Fe ₃ O ₄ (B)	0.70	-0.60	462	21.0
	χ -Fe ₅ C ₂ (I)	0.35	0.05	252	8.2
	χ -Fe ₅ C ₂ (II)	0.28	-0.05	220	24.0
	χ -Fe ₅ C ₂ (III)	0.30	0.00	118	10.6
	θ -Fe ₃ C (a)	0.25	0.25	260	2.4
	θ -Fe ₃ C (b)	0.40	0.30	243	2.5

References and Notes

1. H. M. Torres Galvis, K. P. de Jong, Catalysts for production of lower olefins from synthesis gas: A review. *ACS Catal.* **3**, 2130–2149 (2013). [doi:10.1021/cs4003436](https://doi.org/10.1021/cs4003436)
2. W. Zhou, K. Cheng, J. Kang, C. Zhou, V. Subramanian, Q. Zhang, Y. Wang, New horizon in C1 chemistry: Breaking the selectivity limitation in transformation of syngas and hydrogenation of CO₂ into hydrocarbon chemicals and fuels. *Chem. Soc. Rev.* **48**, 3193–3228 (2019). [doi:10.1039/C8CS00502H](https://doi.org/10.1039/C8CS00502H) [Medline](#)
3. Q. H. Zhang, J. C. Kang, Y. Wang, Development of novel catalysts for Fischer-Tropsch synthesis: Tuning the product selectivity. *ChemCatChem* **2**, 1030–1058 (2010). [doi:10.1002/cctc.201000071](https://doi.org/10.1002/cctc.201000071)
4. J. Bao, G. H. Yang, Y. Yoneyama, N. Tsubaki, Significant advances in C1 catalysis: Highly efficient catalysts and catalytic reactions. *ACS Catal.* **9**, 3026–3053 (2019). [doi:10.1021/acscatal.8b03924](https://doi.org/10.1021/acscatal.8b03924)
5. F. Jiao, J. Li, X. Pan, J. Xiao, H. Li, H. Ma, M. Wei, Y. Pan, Z. Zhou, M. Li, S. Miao, J. Li, Y. Zhu, D. Xiao, T. He, J. Yang, F. Qi, Q. Fu, X. Bao, Selective conversion of syngas to light olefins. *Science* **351**, 1065–1068 (2016). [doi:10.1126/science.aaf1835](https://doi.org/10.1126/science.aaf1835) [Medline](#)
6. Y. F. Zhu, X. Pan, F. Jiao, J. Li, J. Yang, M. Ding, Y. Han, Z. Liu, X. Bao, Role of manganese oxide in syngas conversion to light olefins. *ACS Catal.* **7**, 2800–2804 (2017). [doi:10.1021/acscatal.7b00221](https://doi.org/10.1021/acscatal.7b00221)
7. K. Cheng, B. Gu, X. Liu, J. Kang, Q. Zhang, Y. Wang, Direct and highly selective conversion of synthesis gas into lower olefins: Design of a bifunctional catalyst combining methanol synthesis and carbon-carbon coupling. *Angew. Chem. Int. Ed.* **55**, 4725–4728 (2016). [doi:10.1002/anie.201601208](https://doi.org/10.1002/anie.201601208) [Medline](#)
8. L. Zhong, F. Yu, Y. An, Y. Zhao, Y. Sun, Z. Li, T. Lin, Y. Lin, X. Qi, Y. Dai, L. Gu, J. Hu, S. Jin, Q. Shen, H. Wang, Cobalt carbide nanoprisms for direct production of lower olefins from syngas. *Nature* **538**, 84–87 (2016). [doi:10.1038/nature19786](https://doi.org/10.1038/nature19786) [Medline](#)
9. H. M. Torres Galvis, J. H. Bitter, C. B. Khare, M. Ruitenbeek, A. I. Dugulan, K. P. de Jong, Supported iron nanoparticles as catalysts for sustainable production of lower olefins. *Science* **335**, 835–838 (2012). [doi:10.1126/science.1215614](https://doi.org/10.1126/science.1215614) [Medline](#)
10. J. Z. Lu, L. Yang, B. Xu, Q. Wu, D. Zhang, S. Yuan, Y. Zhai, X. Wang, Y. Fan, Z. Hu, Promotion effects of nitrogen doping into carbon nanotubes on supported iron Fischer–Tropsch catalysts for lower olefins. *ACS Catal.* **4**, 613–621 (2014). [doi:10.1021/cs400931z](https://doi.org/10.1021/cs400931z)
11. P. Zhai, C. Xu, R. Gao, X. Liu, M. Li, W. Li, X. Fu, C. Jia, J. Xie, M. Zhao, X. Wang, Y.-W. Li, Q. Zhang, X.-D. Wen, D. Ma, Highly tunable selectivity for syngas-derived alkenes over zinc and sodium-modulated Fe₅ C₂ catalyst. *Angew. Chem. Int. Ed.* **55**, 9902–9907 (2016). [doi:10.1002/anie.201603556](https://doi.org/10.1002/anie.201603556) [Medline](#)
12. X. Chen, D. Deng, X. Pan, Y. Hu, X. Bao, N-doped graphene as an electron donor of iron catalysts for CO hydrogenation to light olefins. *Chem. Commun.* **51**, 217–220 (2015). [doi:10.1039/C4CC06600F](https://doi.org/10.1039/C4CC06600F) [Medline](#)

13. Y. Liu, J. F. Chen, J. Bao, Y. Zhang, Manganese-modified Fe₃O₄ microsphere catalyst with effective active phase of forming light olefins from syngas. *ACS Catal.* **5**, 3905–3909 (2015). [doi:10.1021/acscatal.5b00492](https://doi.org/10.1021/acscatal.5b00492)
14. J. Wang, Y. Xu, G. Ma, J. Lin, H. Wang, C. Zhang, M. Ding, Directly converting syngas to linear α -olefins over core-shell Fe₃O₄@MnO₂ catalysts. *ACS Appl. Mater. Interfaces* **10**, 43578–43587 (2018). [doi:10.1021/acsami.8b11820](https://doi.org/10.1021/acsami.8b11820) [Medline](#)
15. V. P. Santos, T. A. Wezendonk, J. J. D. Jaén, A. I. Dugulan, M. A. Nasalevich, H.-U. Islam, A. Chojecki, S. Sartipi, X. Sun, A. A. Hakeem, A. C. J. Koeken, M. Ruitenbeek, T. Davidian, G. R. Meima, G. Sankar, F. Kapteijn, M. Makkee, J. Gascon, Metal organic framework-mediated synthesis of highly active and stable Fischer-Tropsch catalysts. *Nat. Commun.* **6**, 6451 (2015). [doi:10.1038/ncomms7451](https://doi.org/10.1038/ncomms7451) [Medline](#)
16. O. Zhuo, L. Yang, F. Gao, B. Xu, Q. Wu, Y. Fan, Y. Zhang, Y. Jiang, R. Huang, X. Wang, Z. Hu, Stabilizing the active phase of iron-based Fischer-Tropsch catalysts for lower olefins: Mechanism and strategy. *Chem. Sci.* **10**, 6083–6090 (2019). [doi:10.1039/C9SC01210A](https://doi.org/10.1039/C9SC01210A) [Medline](#)
17. E. Rytter, A. R. Salman, N. E. Tsakoumis, R. Myrstad, J. Yang, S. Lögberg, A. Holmen, M. Rønning, Hydrophobic catalyst support surfaces by silylation of γ -alumina for Co/Re Fischer-Tropsch synthesis. *Catal. Today* **299**, 20–27 (2018). [doi:10.1016/j.cattod.2017.04.031](https://doi.org/10.1016/j.cattod.2017.04.031)
18. X. F. Yu, J. Zhang, X. Wang, Q. Ma, X. Gao, H. Xia, X. Lai, S. Fan, T.-S. Zhao, Fischer-Tropsch synthesis over methyl modified Fe₂O₃@SiO₂ catalysts with low CO₂ selectivity. *Appl. Catal. B* **232**, 420–428 (2018). [doi:10.1016/j.apcatb.2018.03.048](https://doi.org/10.1016/j.apcatb.2018.03.048)
19. P. Wang, W. Chen, F.-K. Chiang, A. I. Dugulan, Y. Song, R. Pestman, K. Zhang, J. Yao, B. Feng, P. Miao, W. Xu, E. J. M. Hensen, Synthesis of stable and low-CO₂ selective ϵ -iron carbide Fischer-Tropsch catalysts. *Sci. Adv.* **4**, eaau2947 (2018). [doi:10.1126/sciadv.aau2947](https://doi.org/10.1126/sciadv.aau2947) [Medline](#)
20. J. Xie, P. P. Paalanen, T. W. van Deelen, B. M. Weckhuysen, M. J. Louwerse, K. P. de Jong, Promoted cobalt metal catalysts suitable for the production of lower olefins from natural gas. *Nat. Commun.* **10**, 167 (2019). [doi:10.1038/s41467-018-08019-7](https://doi.org/10.1038/s41467-018-08019-7) [Medline](#)
21. S. M. Rodulfo-Baechler, S. L. González-Cortés, J. Orozco, V. Sagredo, B. Fontal, A. J. Mora, G. Delgado, Characterization of modified iron catalysts by X-ray diffraction, infrared spectroscopy, magnetic susceptibility and thermogravimetric analysis. *Mater. Lett.* **58**, 2447–2450 (2004). [doi:10.1016/j.matlet.2004.02.032](https://doi.org/10.1016/j.matlet.2004.02.032)
22. L. H. Shi, D. B. Li, B. Hou, Y. H. Sun, Organic modification of SiO₂ and its influence on the properties of Co-based catalysts for Fischer–Tropsch synthesis. *Chin. J. Catal.* **28**, 999–1002 (2007). [doi:10.1016/S1872-2067\(07\)60084-9](https://doi.org/10.1016/S1872-2067(07)60084-9)
23. Y.-S. Li, J. S. Church, A. L. Woodhead, Infrared and Raman spectroscopic studies on iron oxide magnetic nano-particles and their surface modifications. *J. Magn. Magn. Mater.* **324**, 1543–1550 (2012). [doi:10.1016/j.jmmm.2011.11.065](https://doi.org/10.1016/j.jmmm.2011.11.065)
24. E. de Smit, B. M. Weckhuysen, The renaissance of iron-based Fischer-Tropsch synthesis: On the multifaceted catalyst deactivation behaviour. *Chem. Soc. Rev.* **37**, 2758–2781 (2008). [doi:10.1039/b805427d](https://doi.org/10.1039/b805427d) [Medline](#)

25. M. Y. Ding, Y. Yang, B. Wu, Y. Li, T. Wang, L. Ma, Study on reduction and carburization behaviors of iron phases for iron-based Fischer–Tropsch synthesis catalyst. *Appl. Energy* **160**, 982–989 (2015). [doi:10.1016/j.apenergy.2014.12.042](https://doi.org/10.1016/j.apenergy.2014.12.042)
26. C. N. Satterfield, R. T. Hanlon, S. E. Tung, Z. M. Zou, G. C. Papaefthymiou, Effect of water on the iron-catalyzed Fischer–Tropsch synthesis. *Ind. Eng. Chem. Prod. Res. Dev.* **25**, 407–414 (1986). [doi:10.1021/i300023a007](https://doi.org/10.1021/i300023a007)
27. D. B. Bukur, B. Todici, N. Elbashir, Role of water-gas-shift reaction in Fischer–Tropsch synthesis on iron catalysts: A review. *Catal. Today* **275**, 66–75 (2016). [doi:10.1016/j.cattod.2015.11.005](https://doi.org/10.1016/j.cattod.2015.11.005)
28. J. H. Liu, A. Zhang, X. Jiang, M. Liu, J. Zhu, C. Song, X. Guo, Direct transformation of carbon dioxide to value-added hydrocarbons by physical mixtures of Fe₅C₂ and K-modified Al₂O₃. *Ind. Eng. Chem. Res.* **57**, 9120–9126 (2018). [doi:10.1021/acs.iecr.8b02017](https://doi.org/10.1021/acs.iecr.8b02017)
29. Y. Cheng, J. Lin, T. Wu, H. Wang, S. Xie, Y. Pei, S. Yan, M. Qiao, B. Zong, Mg and K dual-decorated Fe-on-reduced graphene oxide for selective catalyzing CO hydrogenation to light olefins with mitigated CO₂ emission and enhanced activity. *Appl. Catal. B* **204**, 475–485 (2017). [doi:10.1016/j.apcatb.2016.11.058](https://doi.org/10.1016/j.apcatb.2016.11.058)
30. S. Krishnamoorthy, A. Li, E. Iglesia, Pathways for CO₂ formation and conversion during Fischer–Tropsch synthesis on iron-based catalysts. *Catal. Lett.* **80**, 77–86 (2002). [doi:10.1023/A:1015382811877](https://doi.org/10.1023/A:1015382811877)
31. S. Li, S. Krishnamoorthy, A. Li, G. D. Meitzner, E. Iglesia, Promoted iron-based catalysts for the Fischer–Tropsch Synthesis: Design, synthesis, site densities, and catalytic properties. *J. Catal.* **206**, 202–217 (2002). [doi:10.1006/jcat.2001.3506](https://doi.org/10.1006/jcat.2001.3506)
32. M. Ojeda, R. Nabar, A. U. Nilekar, A. Ishikawa, M. Mavrikakis, E. Iglesia, CO activation pathways and the mechanism of Fischer–Tropsch synthesis. *J. Catal.* **272**, 287–297 (2010). [doi:10.1016/j.jcat.2010.04.012](https://doi.org/10.1016/j.jcat.2010.04.012)
33. C. Pirola, C. L. Bianchi, A. Di Michele, S. Vitali, V. Ragaini, Fischer Tropsch and Water Gas Shift chemical regimes on supported iron-based catalysts at high metal loading. *Catal. Commun.* **10**, 823–827 (2009). [doi:10.1016/j.catcom.2008.12.006](https://doi.org/10.1016/j.catcom.2008.12.006)
34. G. P. van der Laan, A. A. C. M. Beenackers, Intrinsic kinetics of the gas–solid Fischer–Tropsch and water gas shift reactions over a precipitated iron catalyst. *Appl. Catal. A Gen.* **193**, 39–53 (2000). [doi:10.1016/S0926-860X\(99\)00412-3](https://doi.org/10.1016/S0926-860X(99)00412-3)
35. Y. X. Li, J.-X. Shen, S.-S. Peng, J.-K. Zhang, J. Wu, X.-Q. Liu, L.-B. Sun, Enhancing oxidation resistance of Cu(I) by tailoring microenvironment in zeolites for efficient adsorptive desulfurization. *Nat. Commun.* **11**, 3206 (2020). [doi:10.1038/s41467-020-17042-6](https://doi.org/10.1038/s41467-020-17042-6) [Medline](#)
36. Z. Jin, L. Wang, Q. Hu, L. Zhang, S. Xu, X. Dong, X. Gao, R. Ma, X. Meng, F.-S. Xiao, Hydrophobic zeolite containing titania particles as wettability-selective catalyst for formaldehyde removal. *ACS Catal.* **8**, 5250–5254 (2018). [doi:10.1021/acscatal.8b00732](https://doi.org/10.1021/acscatal.8b00732)
37. Z. Jin, L. Wang, E. Zuidema, K. Mondal, M. Zhang, J. Zhang, C. Wang, X. Meng, H. Yang, C. Mesters, F. S. Xiao, Hydrophobic zeolite modification for in situ peroxide formation

in methane oxidation to methanol. *Science* **367**, 193–197 (2020).
[10.1126/science.aaw1108](https://doi.org/10.1126/science.aaw1108) [Medline](#)

38. W. Y. Wang, W. Zhou, W. Li, X. Xiong, Y. Wang, K. Cheng, J. Kang, Q. Zhang, Y. Wang, In-situ confinement of ultrasmall palladium nanoparticles in silicalite-1 for methane combustion with excellent activity and hydrothermal stability. *Appl. Catal. B* **276**, 119142 (2020). [doi:10.1016/j.apcatb.2020.119142](https://doi.org/10.1016/j.apcatb.2020.119142)
39. Y. F. Xu, J. Liu, G. Ma, J. Wang, J. Lin, H. Wang, C. Zhang, M. Ding, Effect of iron loading on acidity and performance of Fe/HZSM-5 catalyst for direct synthesis of aromatics from syngas. *Fuel* **228**, 1–9 (2018). [doi:10.1016/j.fuel.2018.04.151](https://doi.org/10.1016/j.fuel.2018.04.151)